CSE2301– DATA COMMUNICATION AND COMPUTER NETWORKS

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UNIT II

Network layer – Switching concepts – Circuit switching – Packet switching – IP Addressing –IPV4, IPV6.

OVERVIEW

- **Network Layer**
- **Switching Concepts**
- **Circuit Switching**
- **Packet Switching**
- **Message Switching**
- **IP Addressing**
- **IPV4**
- **IPV6**
- **Routing Protocols**
- **Distance Vector Routing**
- **Link State Routing**

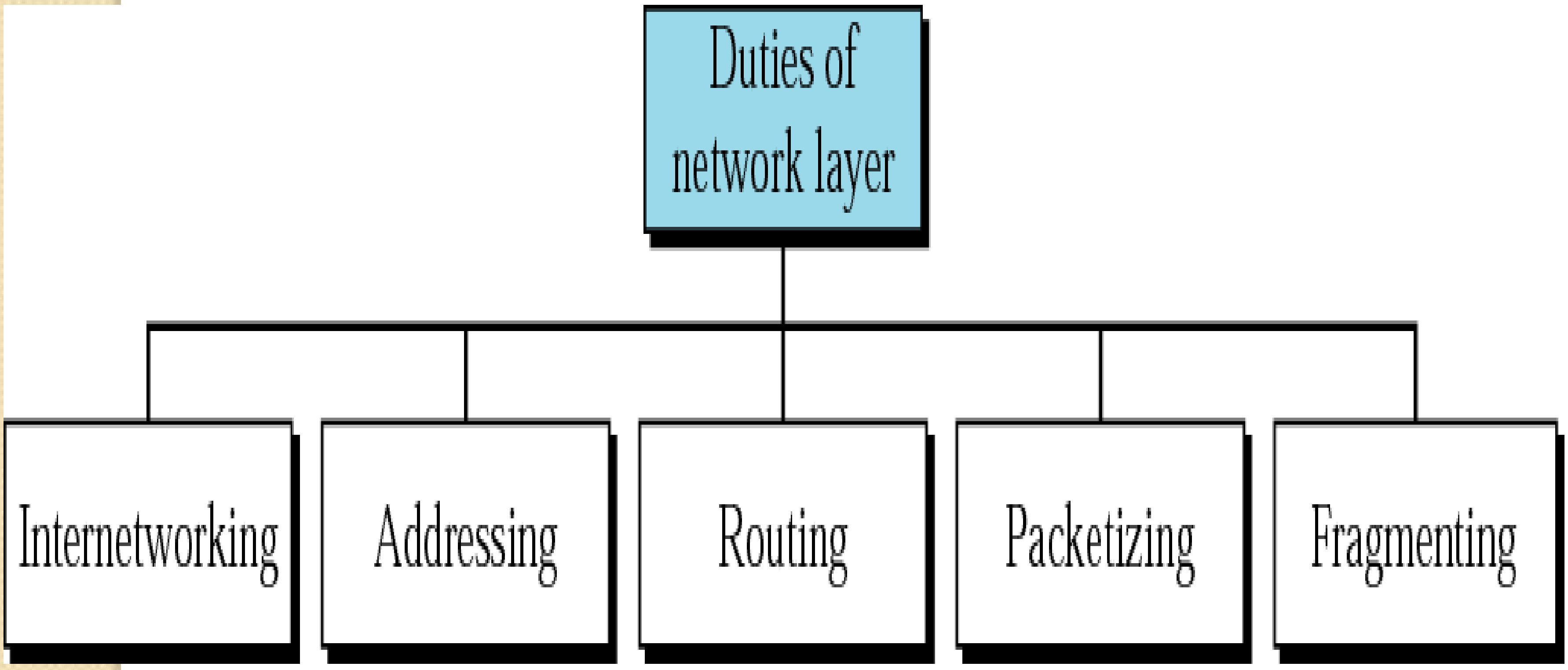
Network Layer

- The Network layer is responsible for the **source-to-destination delivery of a packet** possible across multiple networks.
- It converts **Frames into packets.**

Functions of Network Layer

- **Source-to-Destination delivery of a packet**
- **Logical addressing**
- **Routing**
- **Internetworking**

Network layer Duties



Switching Concepts

- Switches are hardware or software devices used for temporary connection b/w 2 or more devices linked to the switch in network **but not to each another**
- Switches are needed to connect multiple devices for making one-one communication

TYPES:

- Circuit Switching
- Packet Switching
- Message Switching

Figure 14.1 *Switched network*

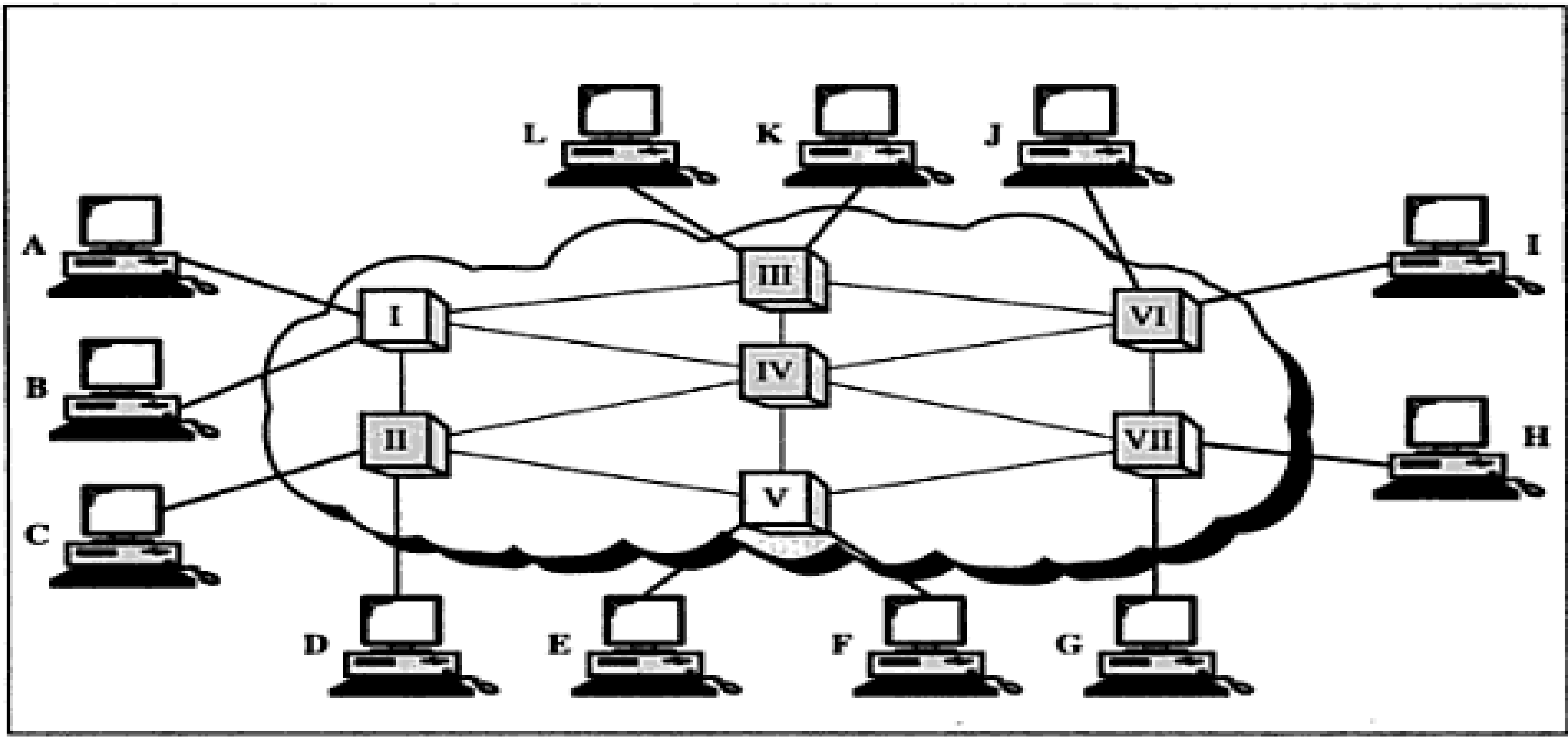


Figure 14.2 *Switching methods*

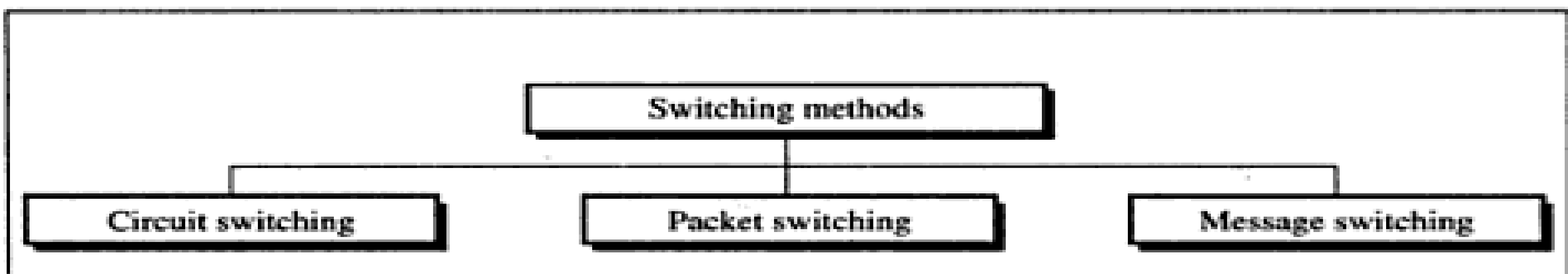


Figure 14.3 *Circuit-switched network*

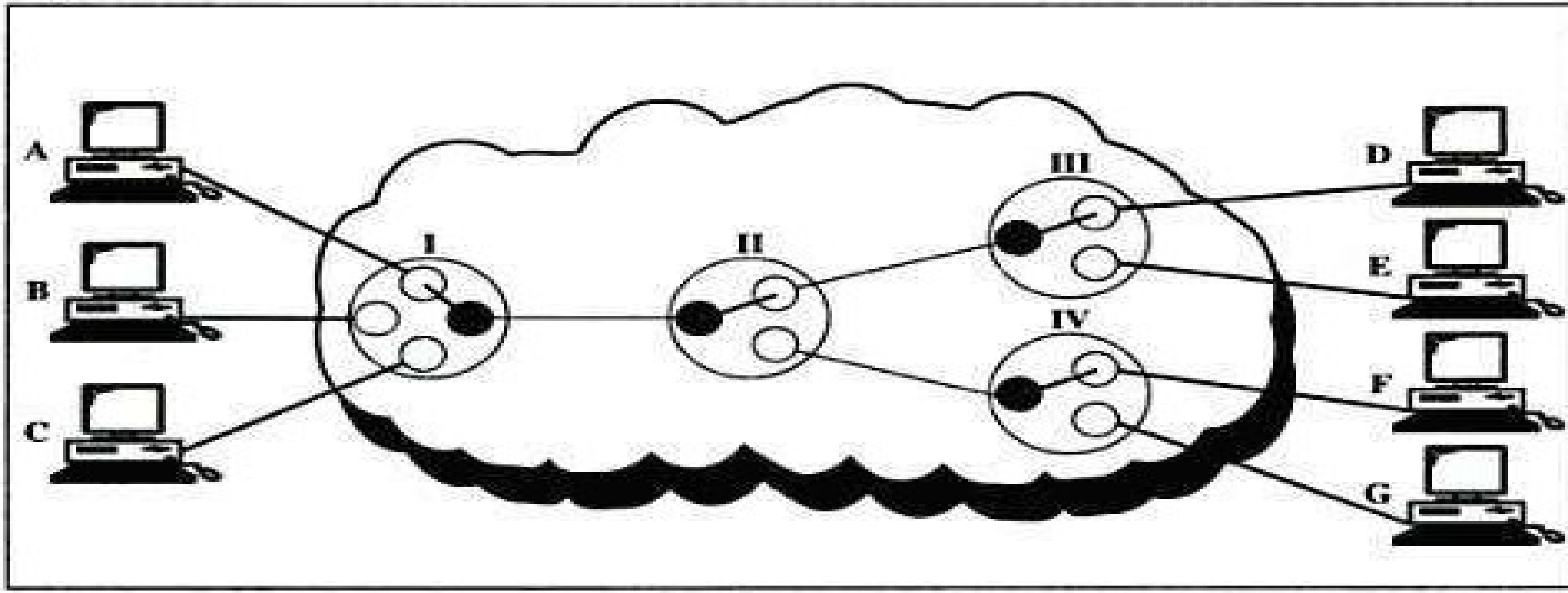


Figure 14.4 *A circuit switch*

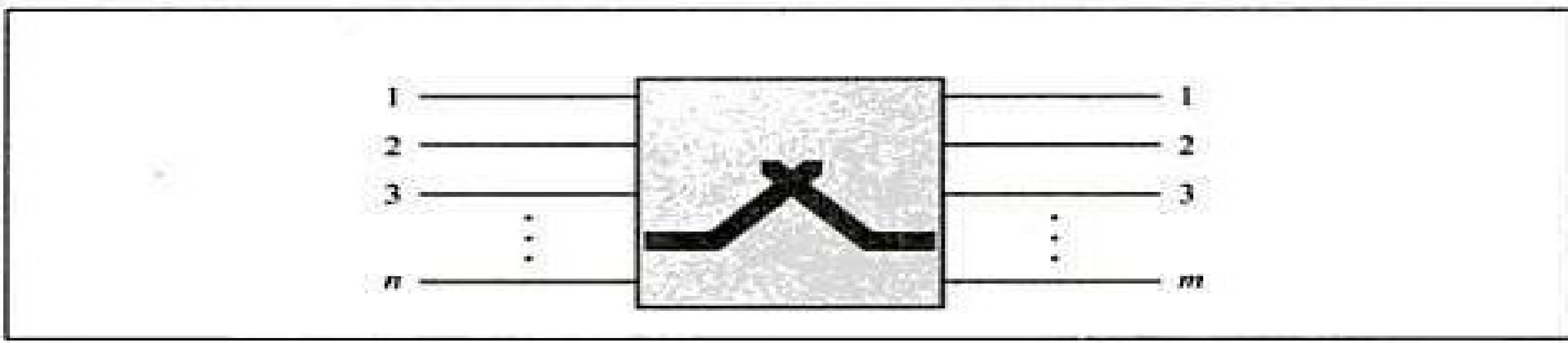
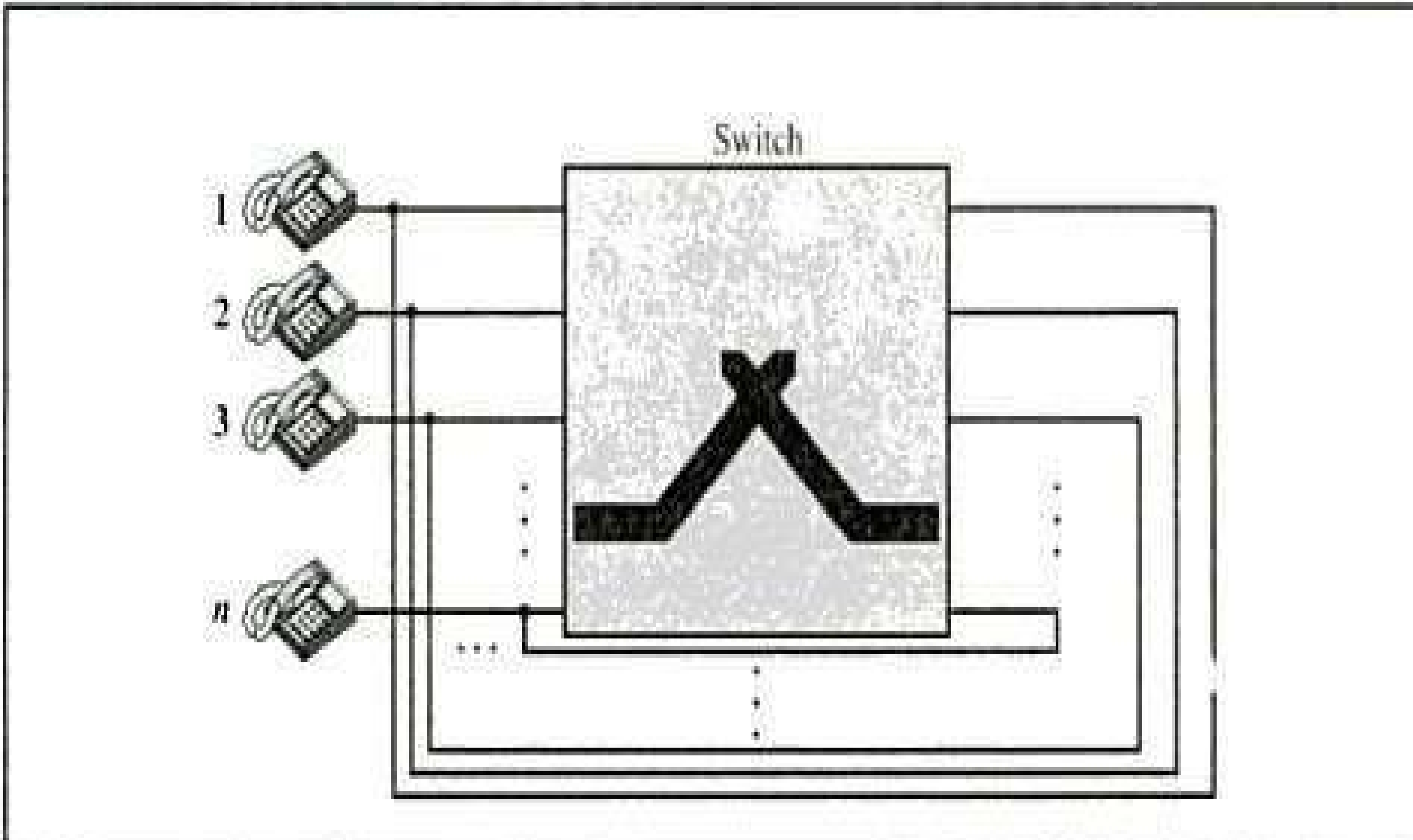


Figure 14.5 *A folded switch*



Space Division Switch

- Path in the circuit are separated from each other
- It is used both in analog and digital communication
- <u>2 Types:</u>
  - **Crossbar switch**
  - **Multistage switch**

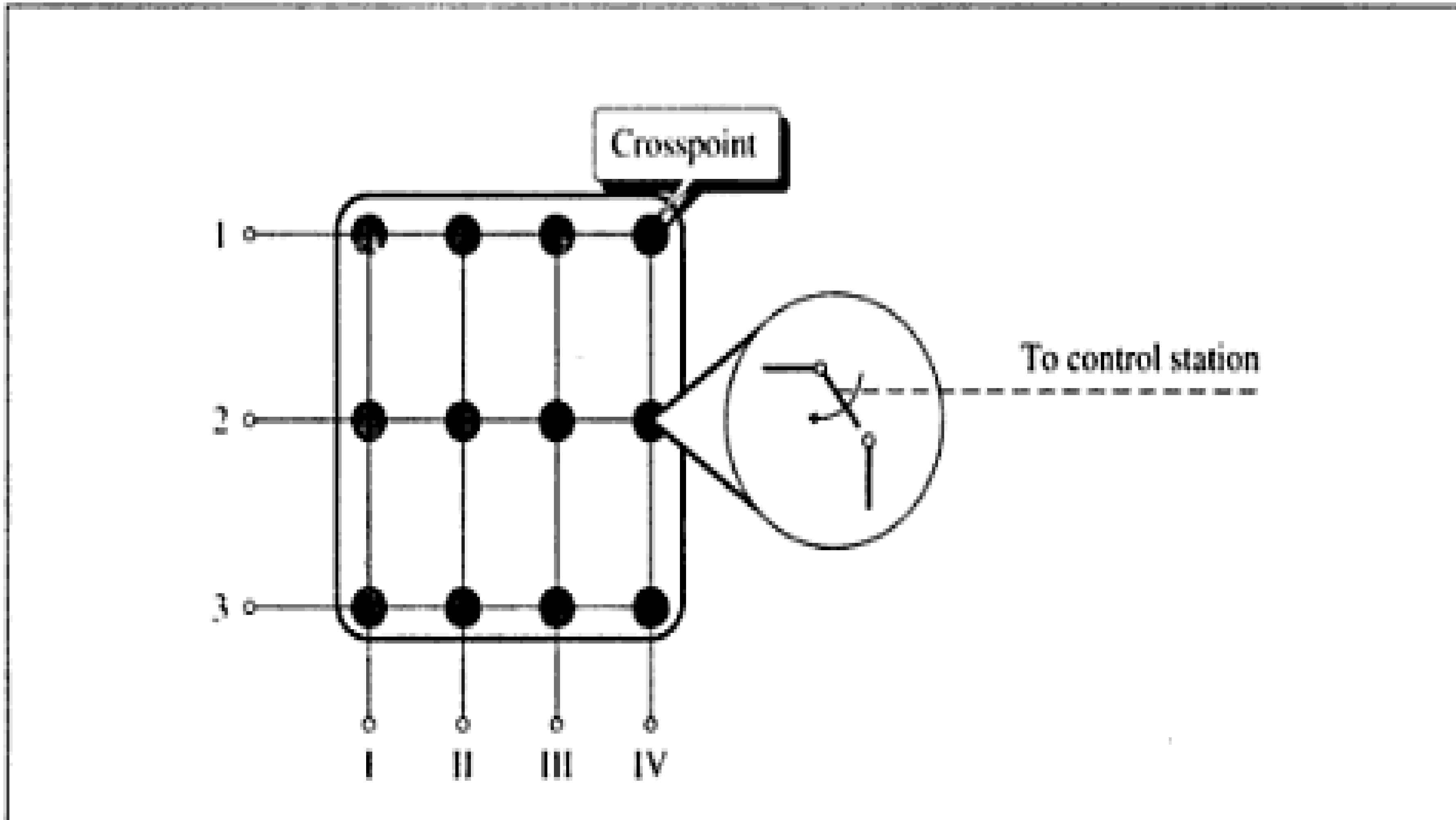
- <u>Crossbar Switch:</u>
  - It connects n inputs to moutputs using cross
  - points

- <u>Limitation:</u>

  More cross points needed(1000 I/P - 1000 O/P requires 1000000 crosspoints)

Crossbar Switch:

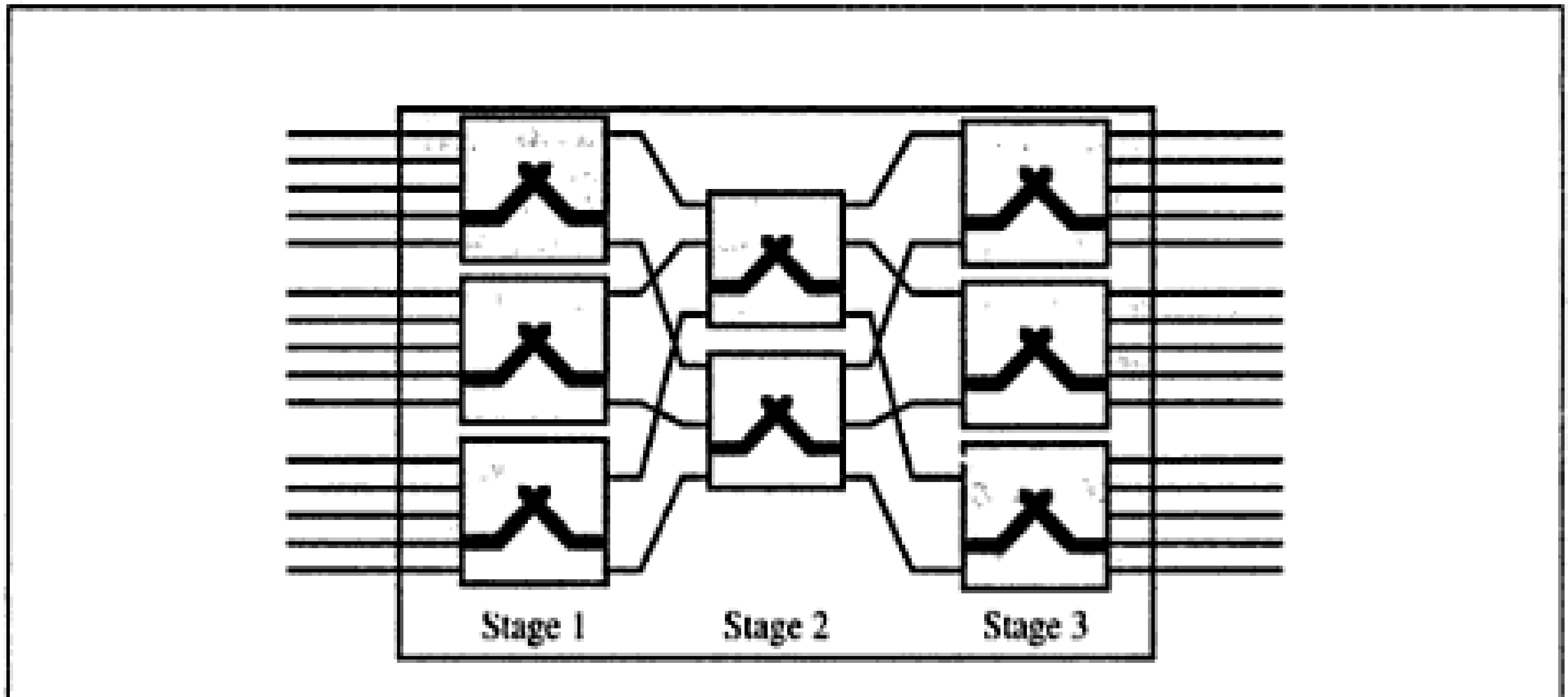
Figure 14.7 *Crossbar switch*



Multistage switch

- Devices are linked to switches ,that are in turn linked to another switches(Hierarchy of switches)

Figure 14.8 *Multistage switch*



Blocking:

- The **reduction in a number of cross points** causes a phenomena called Blocking.
- During heavy traffic one input cannot be connected to output because no path available

Time Division Switches

- It uses time division multiplexing

- 2 methods:
- **Time slot**
- **interchange TDM bus**

Time slot interchange:

- It changes the ordering of the slot based on the desired connection
- It uses RAM to store time
- slot Ex:
- 1->3 2->4 3->1 4->2

TSI

Figure 14.10 *Time-division multiplexing, without and with a time-slot interchange (TSI)*

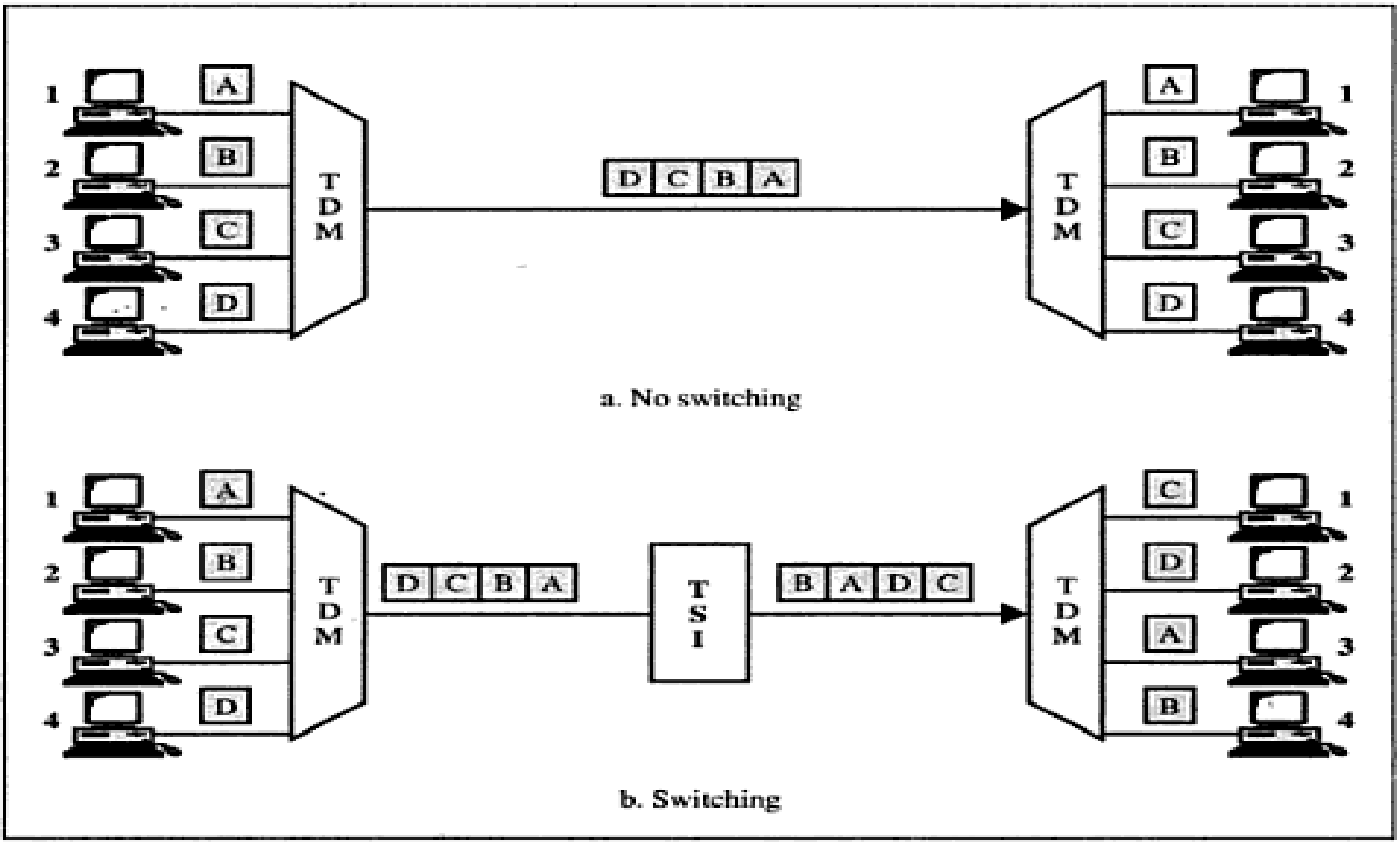
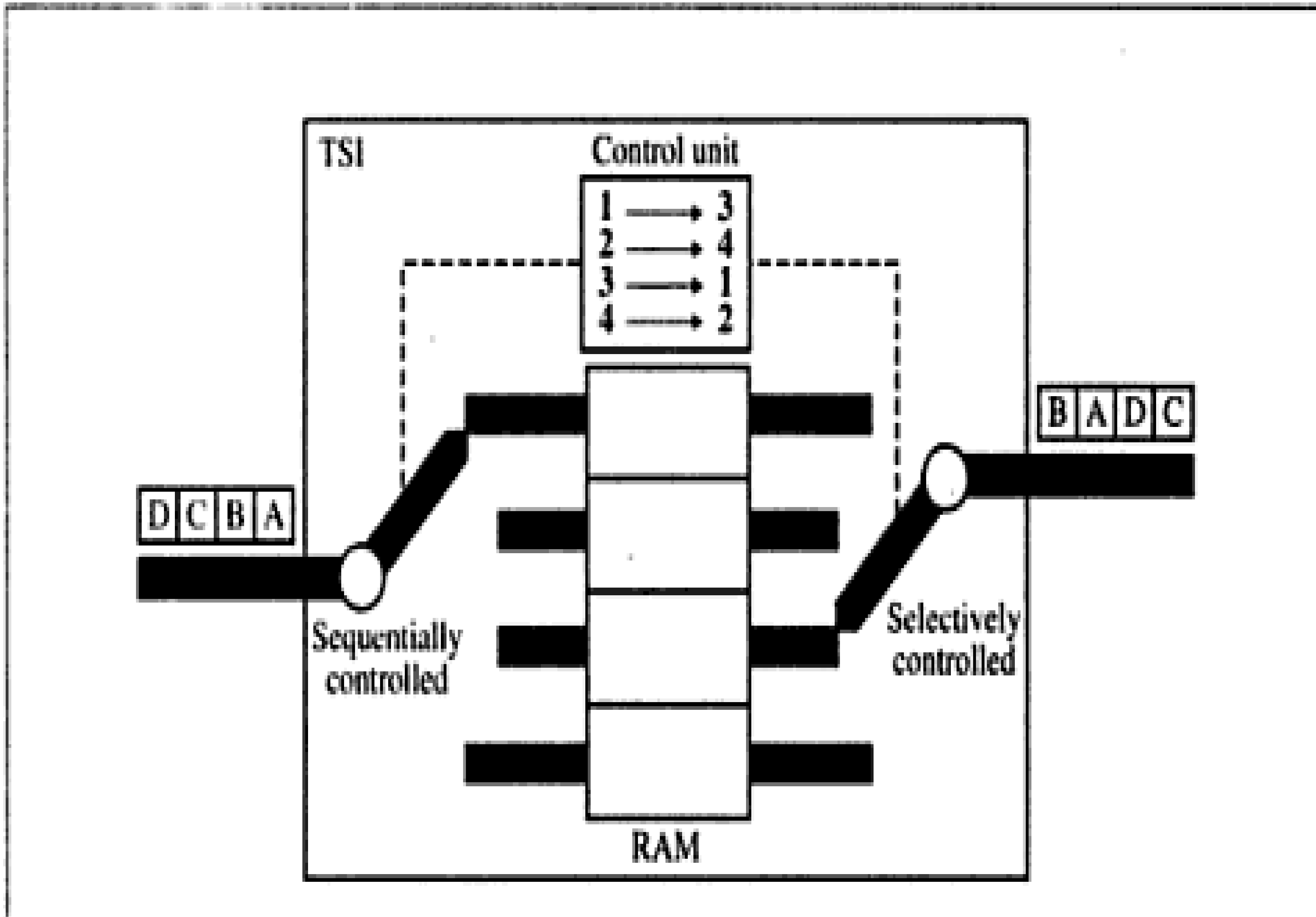


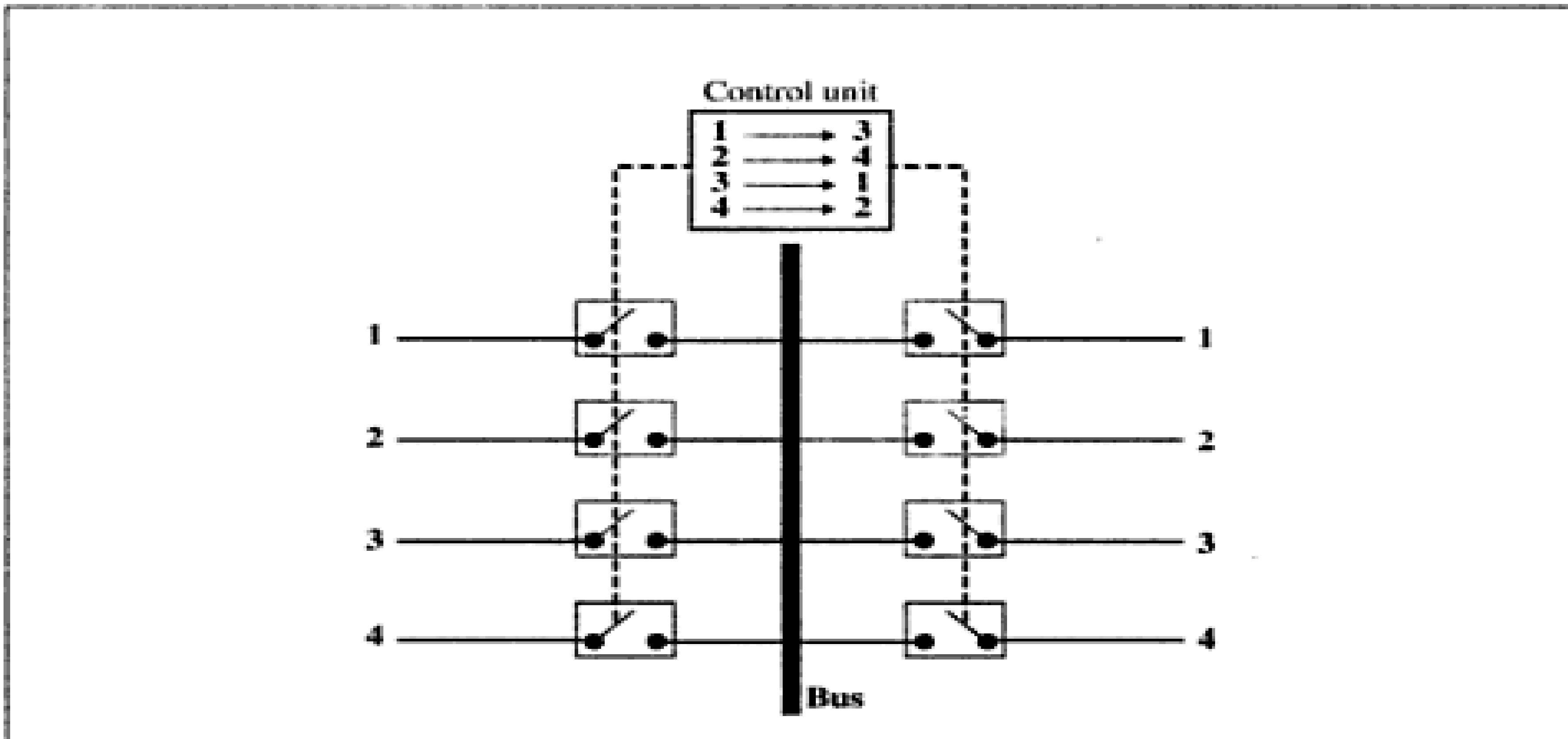
Figure 14.11 *Time-slot interchange*



TDM Bus-Time Division

- **Multiplexing** Here each input and output lines are connected to high speed bus
- Each bus is closed during one of the four time

Figure 14.12 *TDM bus*



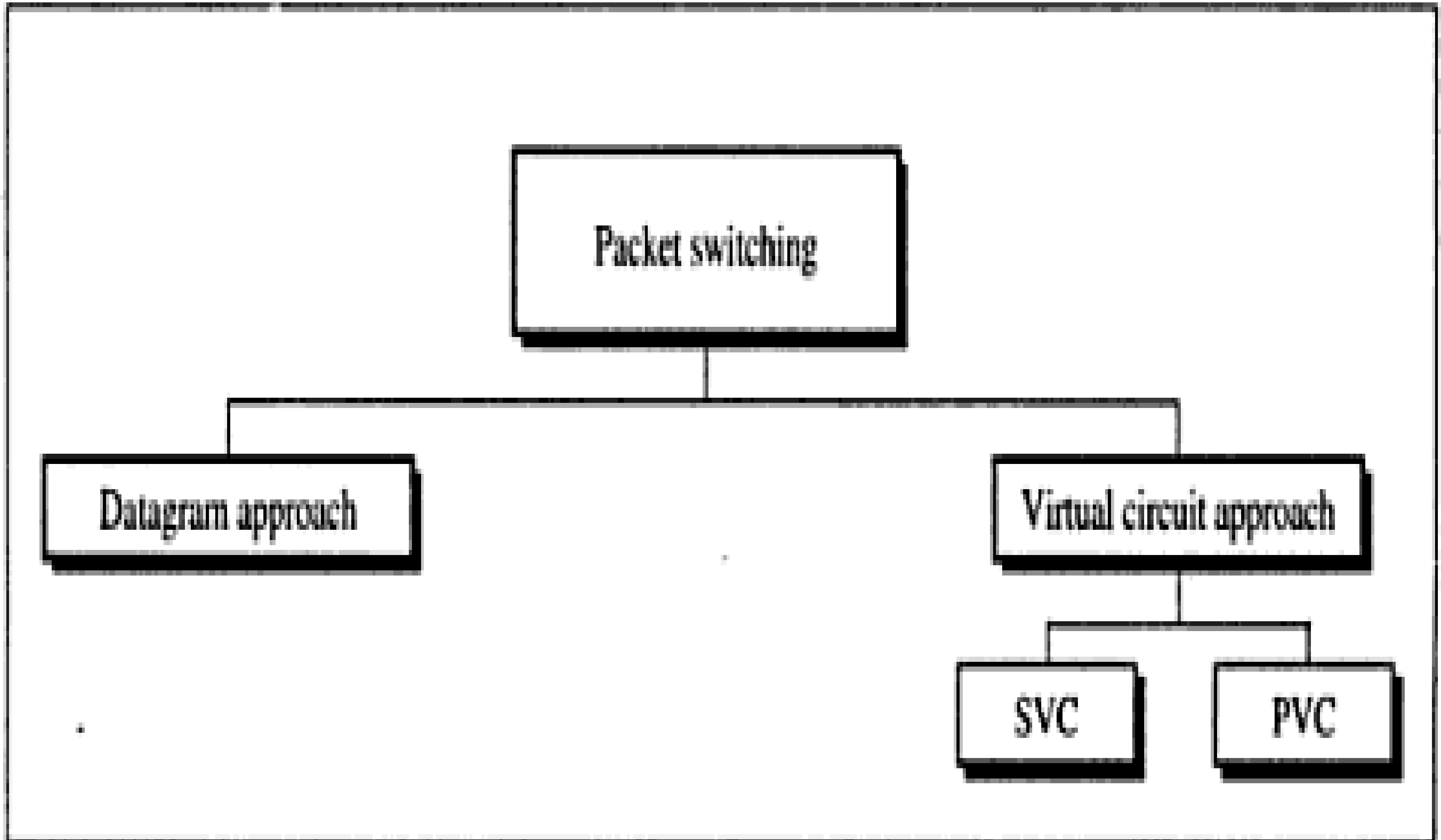
Limitations of Circuit Switching

- It is **specially designed for voice communication** (telephone). **Not suitable for data communication.**
- Once a circuit is established, it remains for duration of the session. It creates dialed(temporary) and leased(Permanent).
- Less data rate because of point to point connection.

Packet switching

- Packet switching is better for data transmission.
- Here data are transmitted through unit of variable length blocks called **packets**.
- Longer transmission are divided into multiple packets.
- Packet length is decided by network.

Figure 14.16 *Packet-switching approaches*



Datagram Approach

- In this approach a message is divided into multiple packets.
- All packets choose various routes and reaches the destination.
- Ordering of packets in destination is done by transport layer.

Figure 14.17 *Datagram approach*

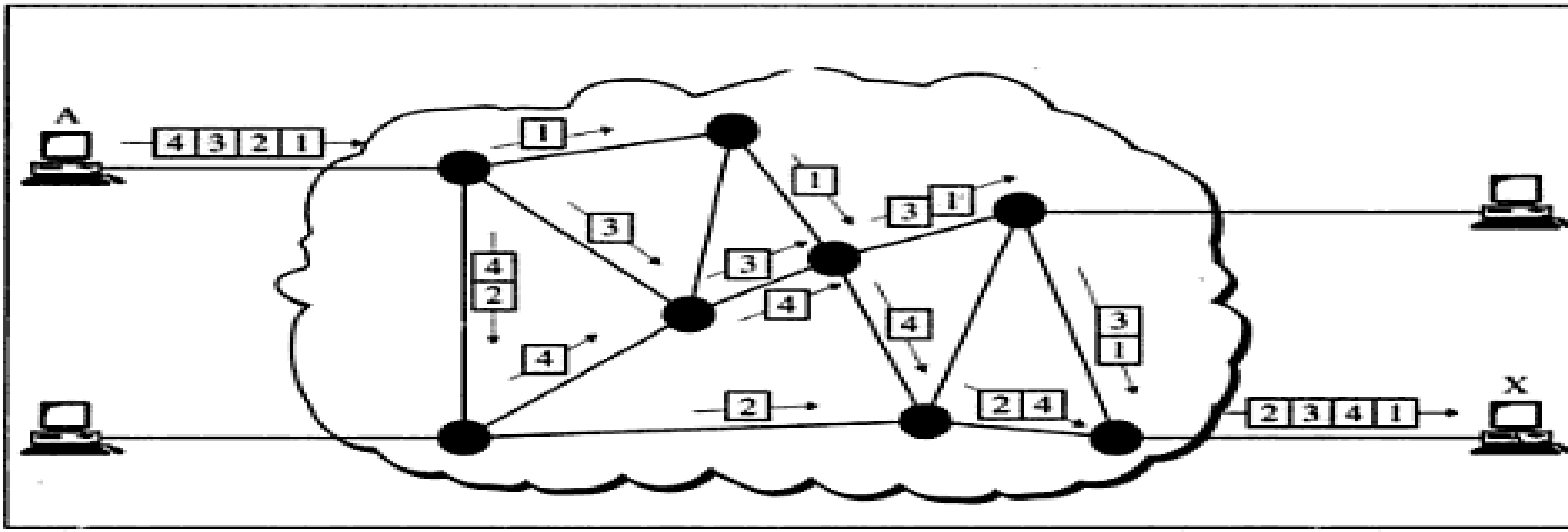
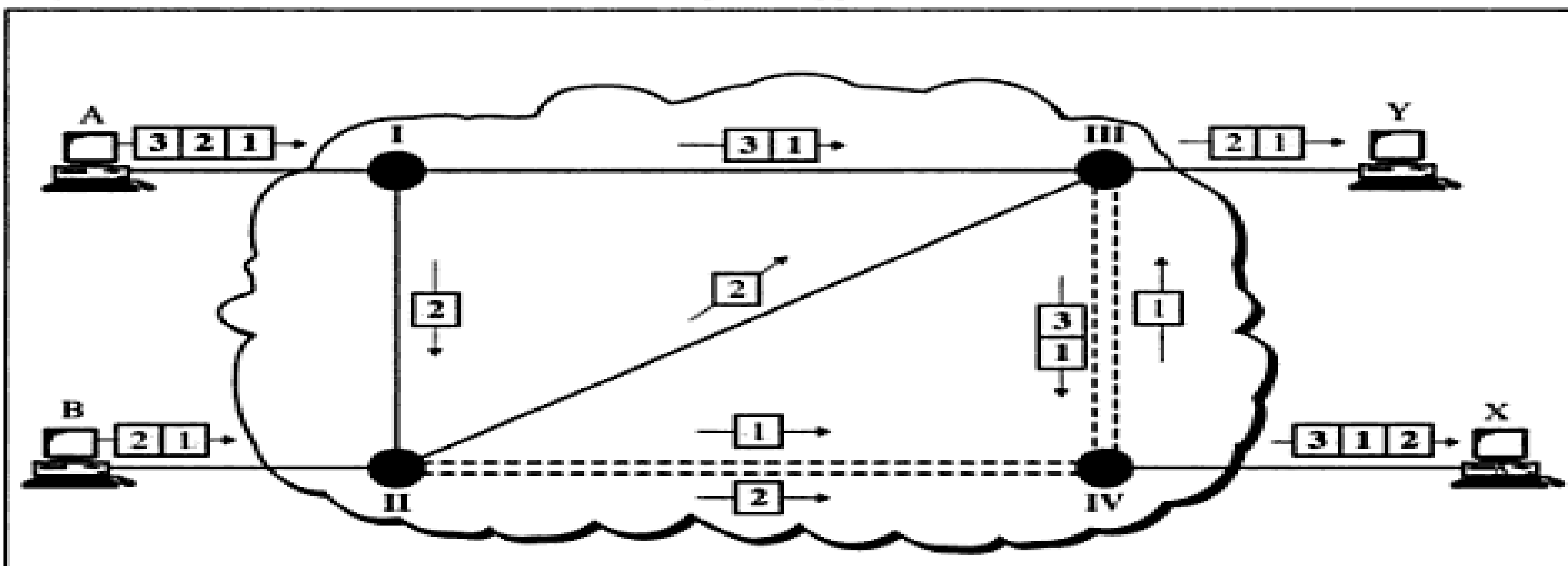


Figure 14.18 *Multiple channels in datagram approach*



Virtual Circuit approach

- It uses single route to send all packets of the message

Two formats:

- Switched virtual circuit
- Permanent virtual circuit

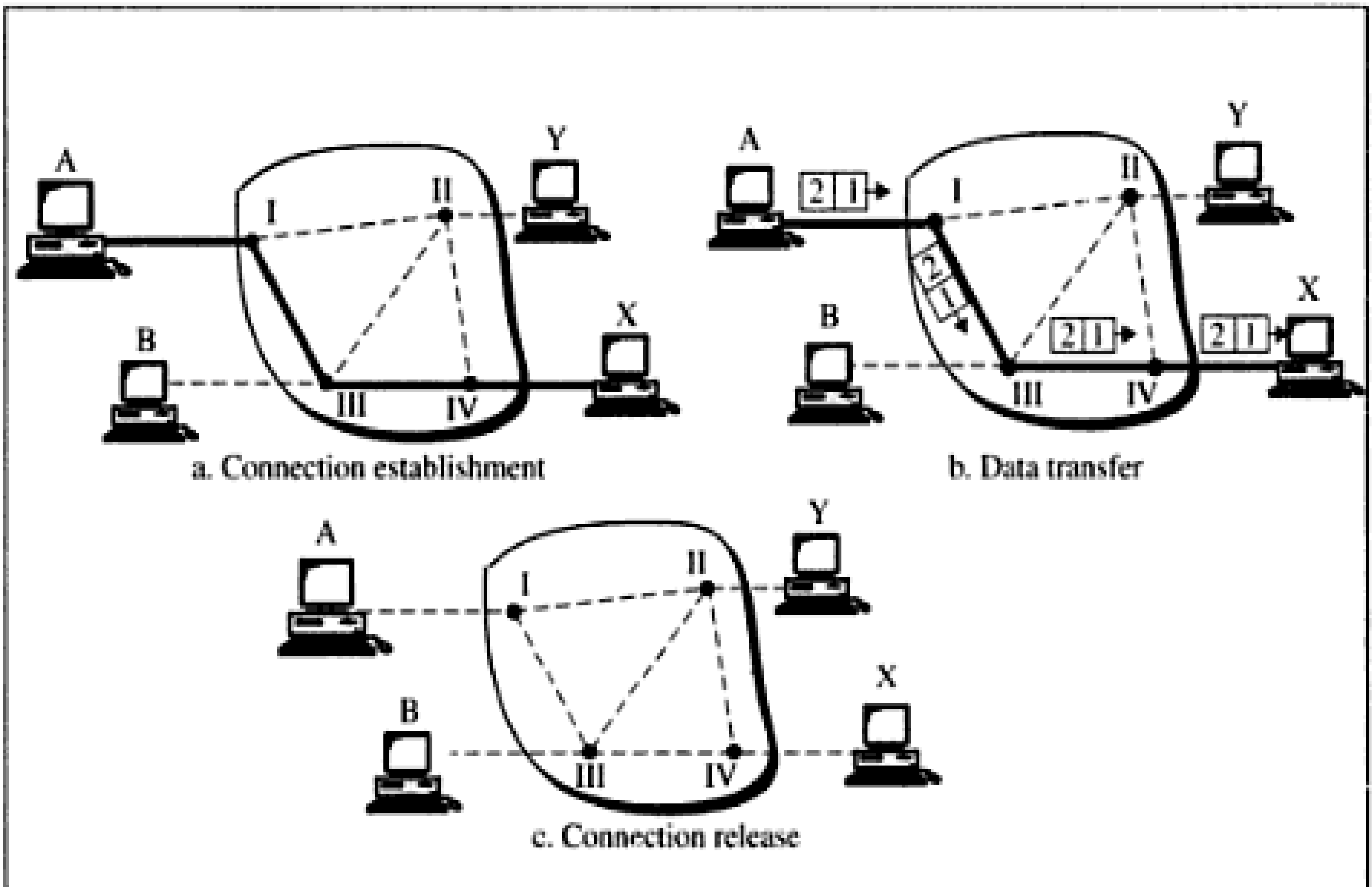
SVC

- Connection is **temporary**
- **Dial-up lines**

During Transmission.

- A connection is established-all packets are sent – proper ACK- Connection is terminated

Figure 14.19 *Switched virtual circuit (SVC)*



PVC

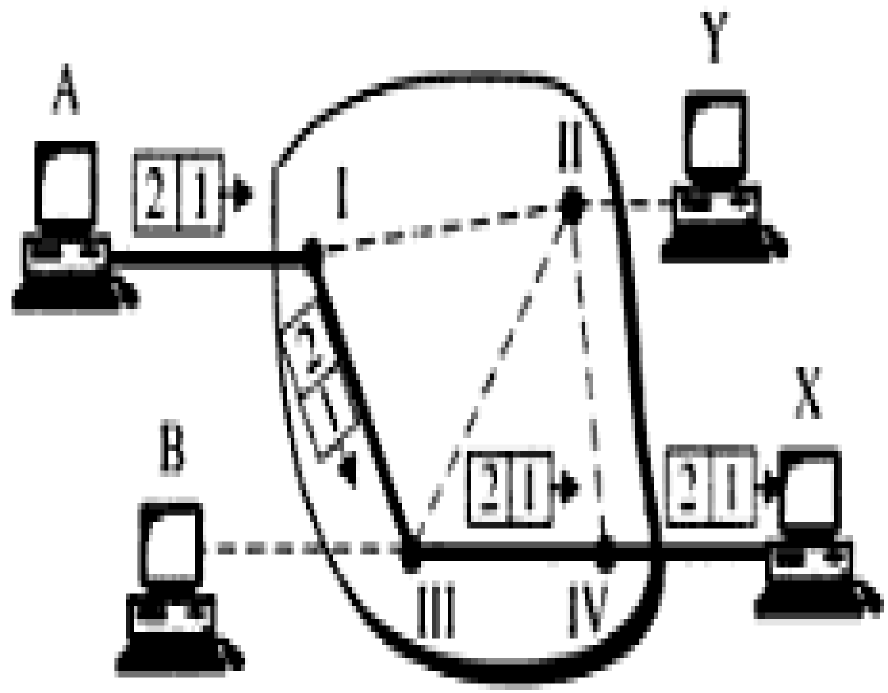
- Connection is **permanent.**
- Circuit is dedicated for two users, Noone else can use the line when communication takes place.
- It always gets the same route.
- **Leased lines.**

During Transmission.

No connection establishment or termination

PVC

Figure 14.20 *Permanent virtual circuit (PVC)*



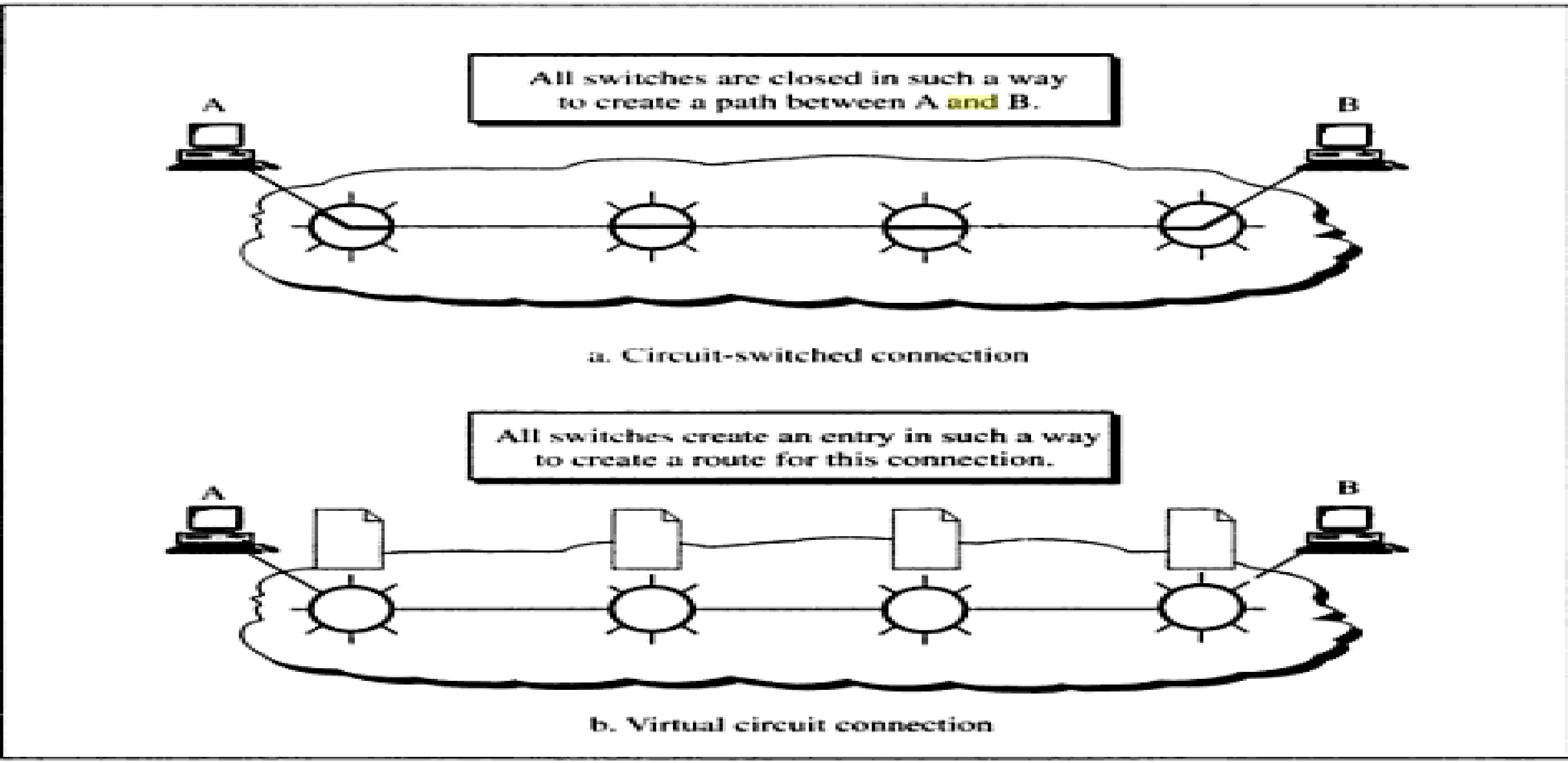
Permanent connection for the duration of the lease

Circuit switched Vs Virtual Circuit

Path Vs Route:

- Circuit switched->Path
- Virtual Circuit->Route

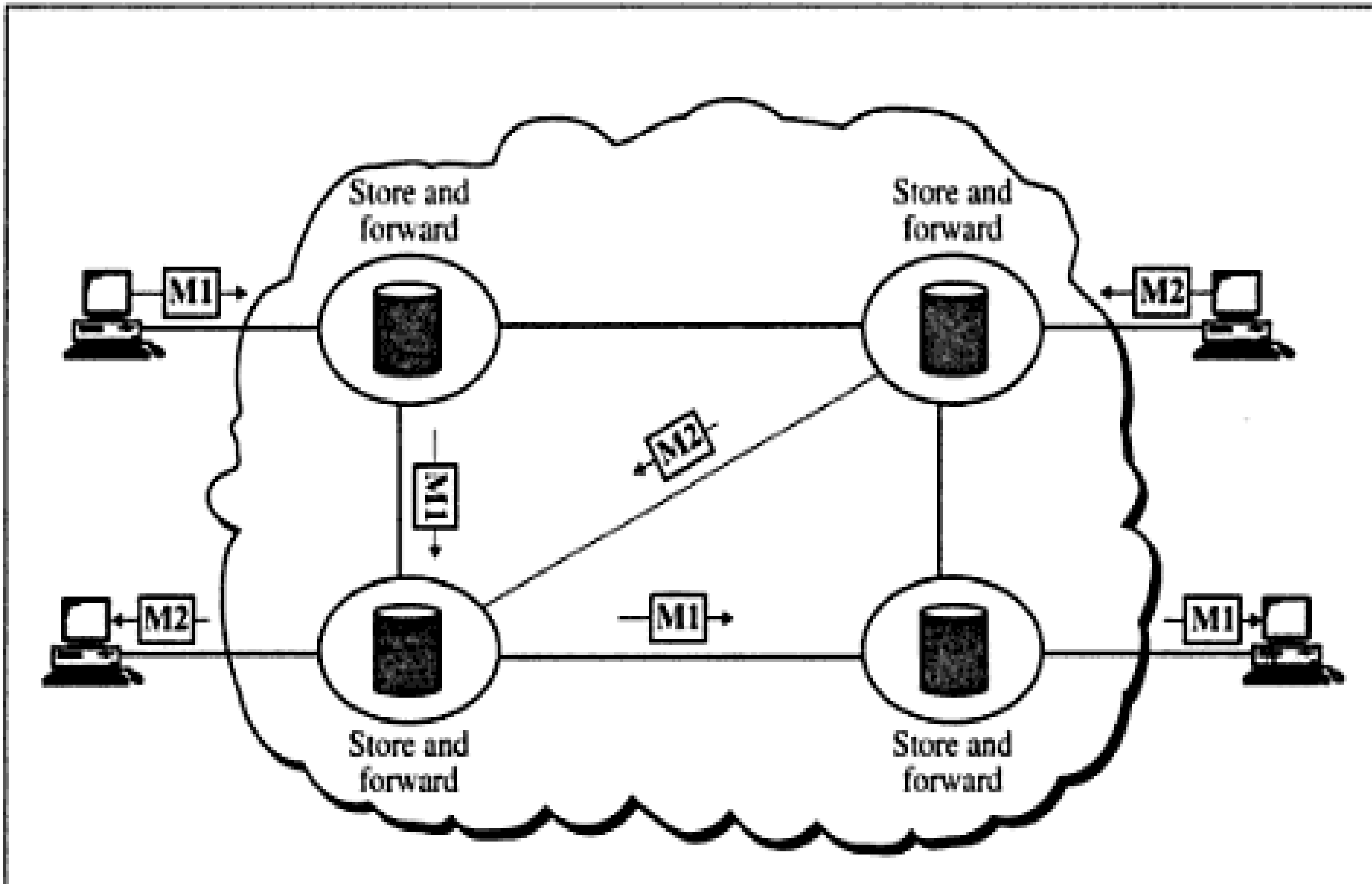
Figure 14.21 *Path versus route*



Message Switching

- It uses a mechanism called **store and forward**
- Here a message is received and stored until a appropriate route is free, then sends along.
- Message switching- uses secondary storage(Disk)
- Packet switching – uses primary storage(RAM)

Figure 14.23 *Message switching*



Routers

- The routers decide which route is bestamong many routes in a particular transmission.
- Routers are like stations on the network

<u>Routing concepts:</u>

- Least cost
- routing:
- Cheaper
- Shortest path(using small number of relays or **hops**. Hop-count ->Number of relays

Non - Adaptive Routing

- In some routing protocols , once a pathway to a destination is selected ,the router sends all packets in that way.
- ## Adaptive Routing:
- The router may select new route for each packet.

Packet Life Time (or)Time to Live:

The problem created by looping or bouncing is avoided by destroying the packet without looping, New packet is retransmitted

Routing Algorithms or Routing Protocols

- To route the packet with optimal cost many routing algorithms are used to Calculating the shortest path between 2 routers

1. Distance Vector Routing
2. Link State Routing

Distance vector Routing

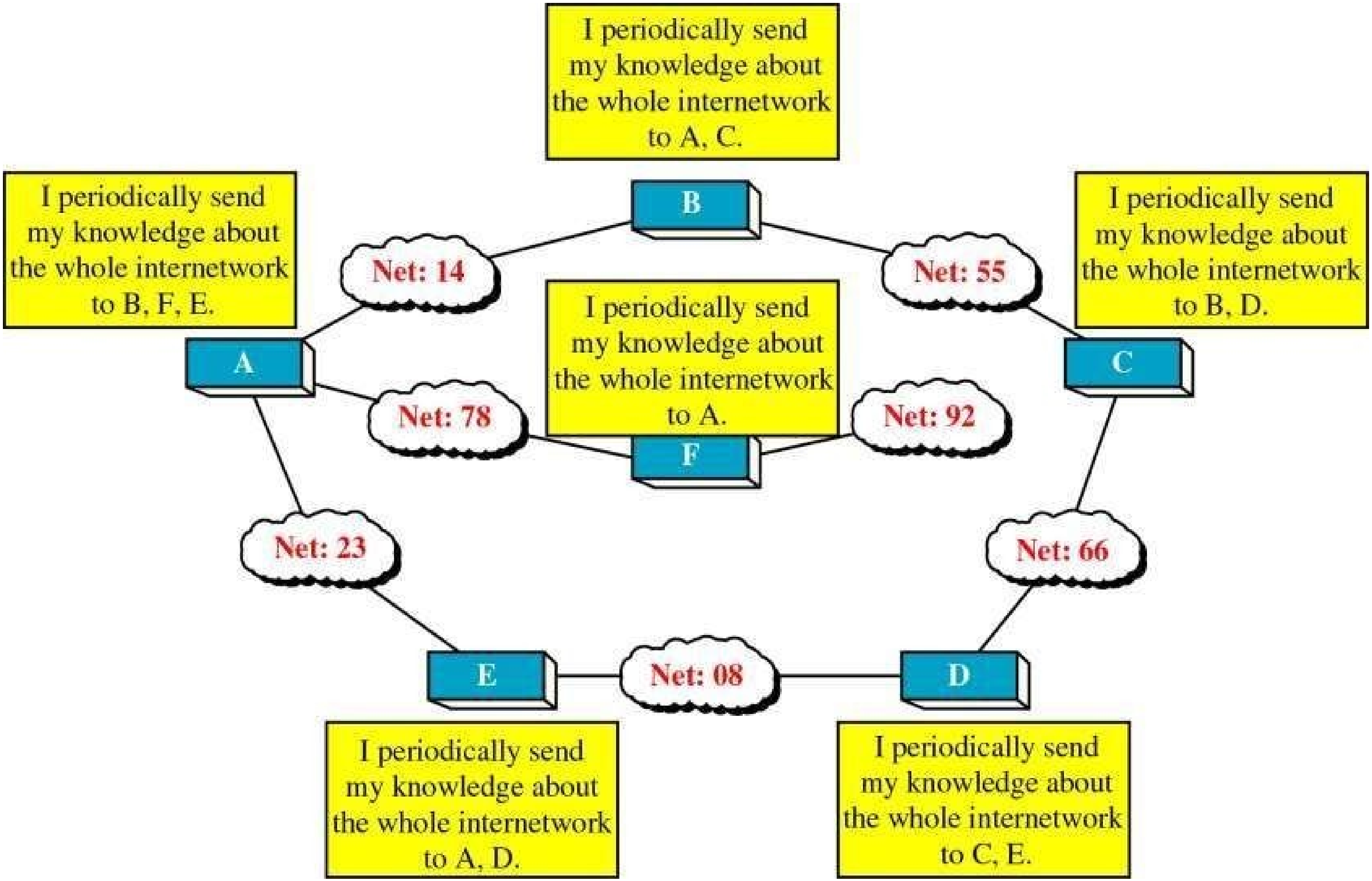
Def:

- Each router periodically shares its knowledge about the entire network with its neighbor.
- It is represented by graph.

<u>Key Works</u>:

- Each router shares its knowledge about the entire network to neighbors.
- Routing only to the directly linked routers.
- Information sharing at regular interval(each 30 seconds).

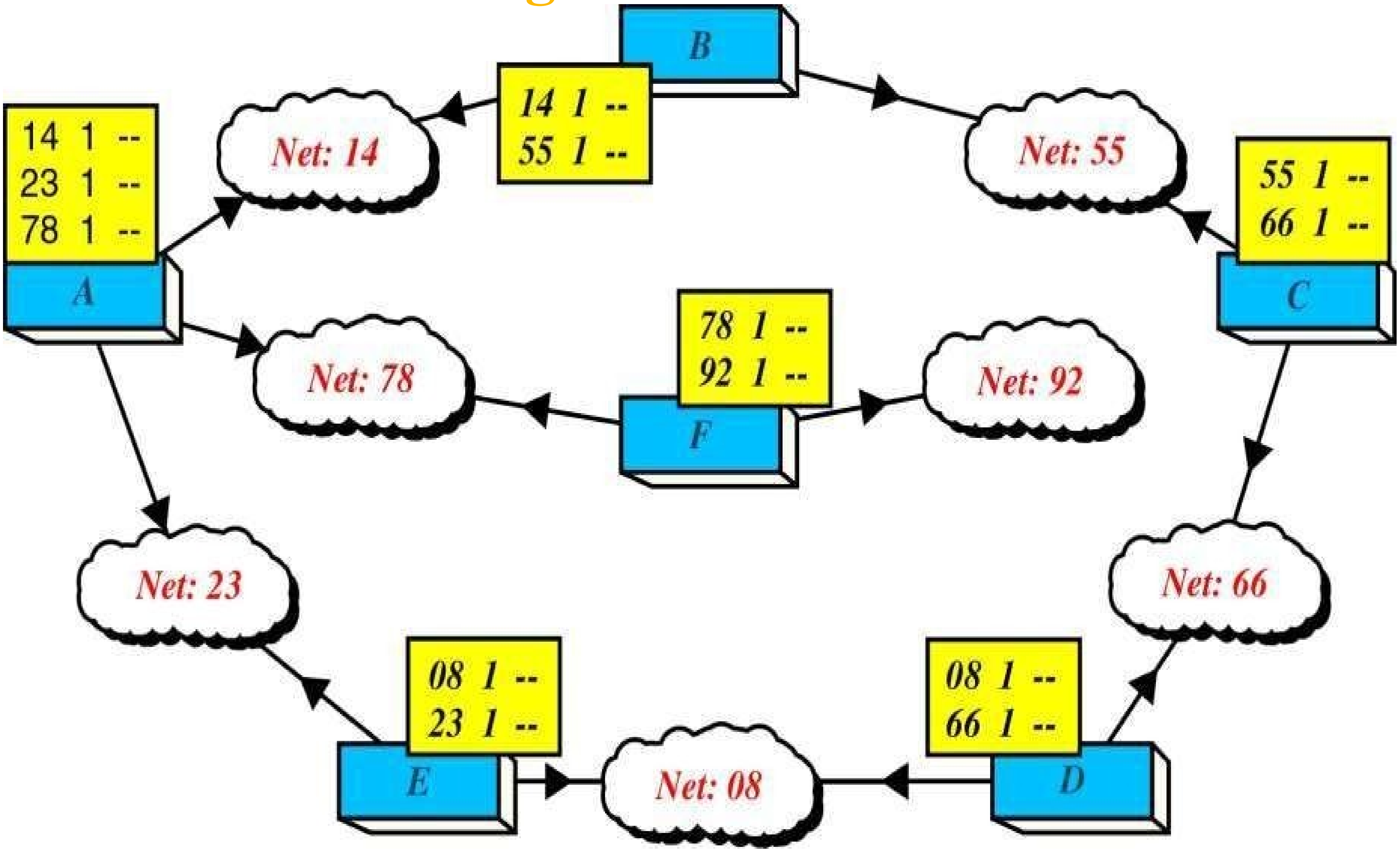
The Concept of Distance Vector Routing



Distance Vector Routing Table

Network ID	Cost	Next Hop
. . . .	. . . .	. . . .
. . . .	. . . .	. . . .
. . . .	. . . .	. . . .
. . . .	. . . .	. . . .
. . . .	. . . .	. . . .

Routing Table Distribution



Link State Routing

Def:

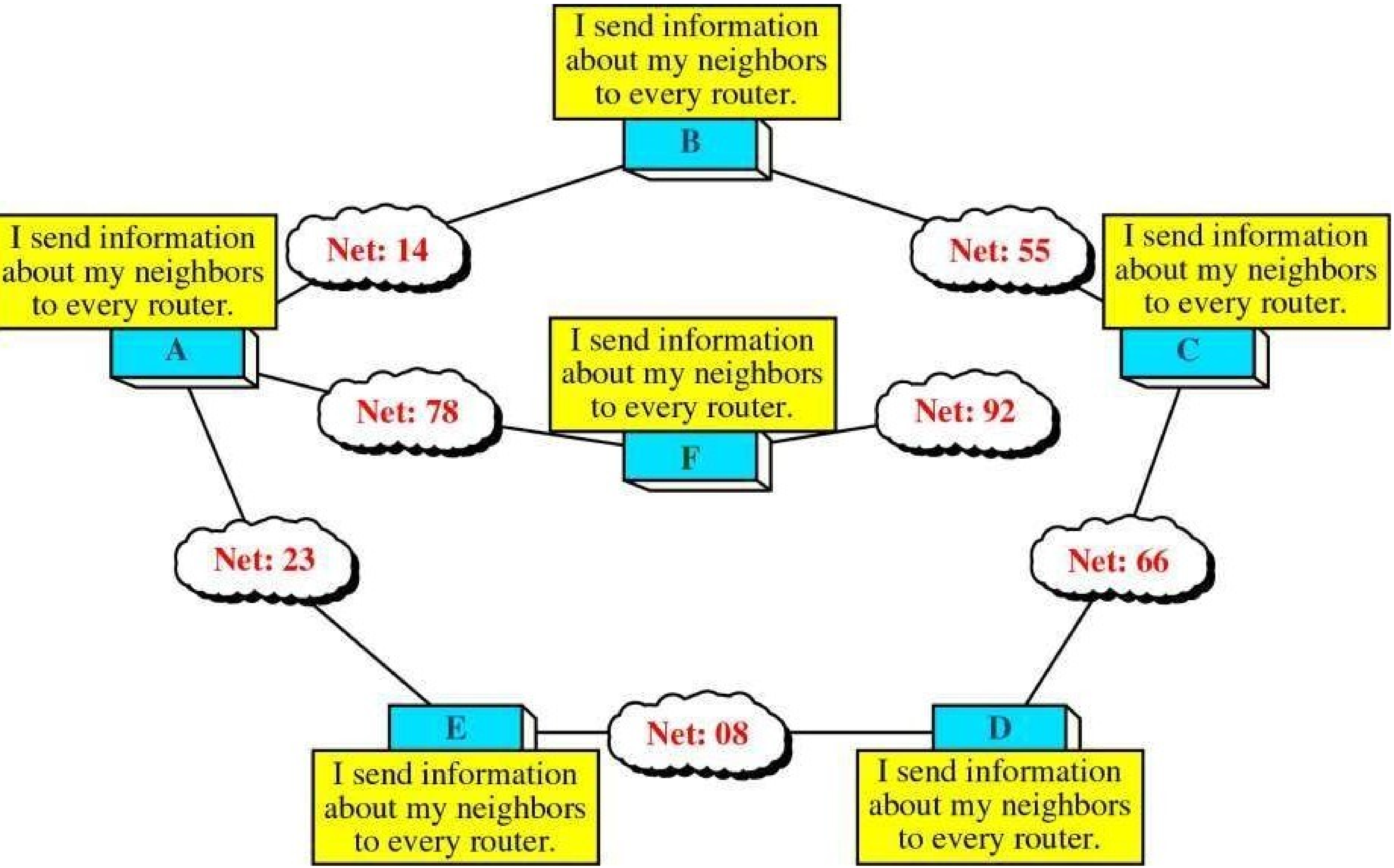
- Each router shares its knowledge of it neighborhood with all routers in the internetwork.
- It is represented by directed graph with weight.

Key work:

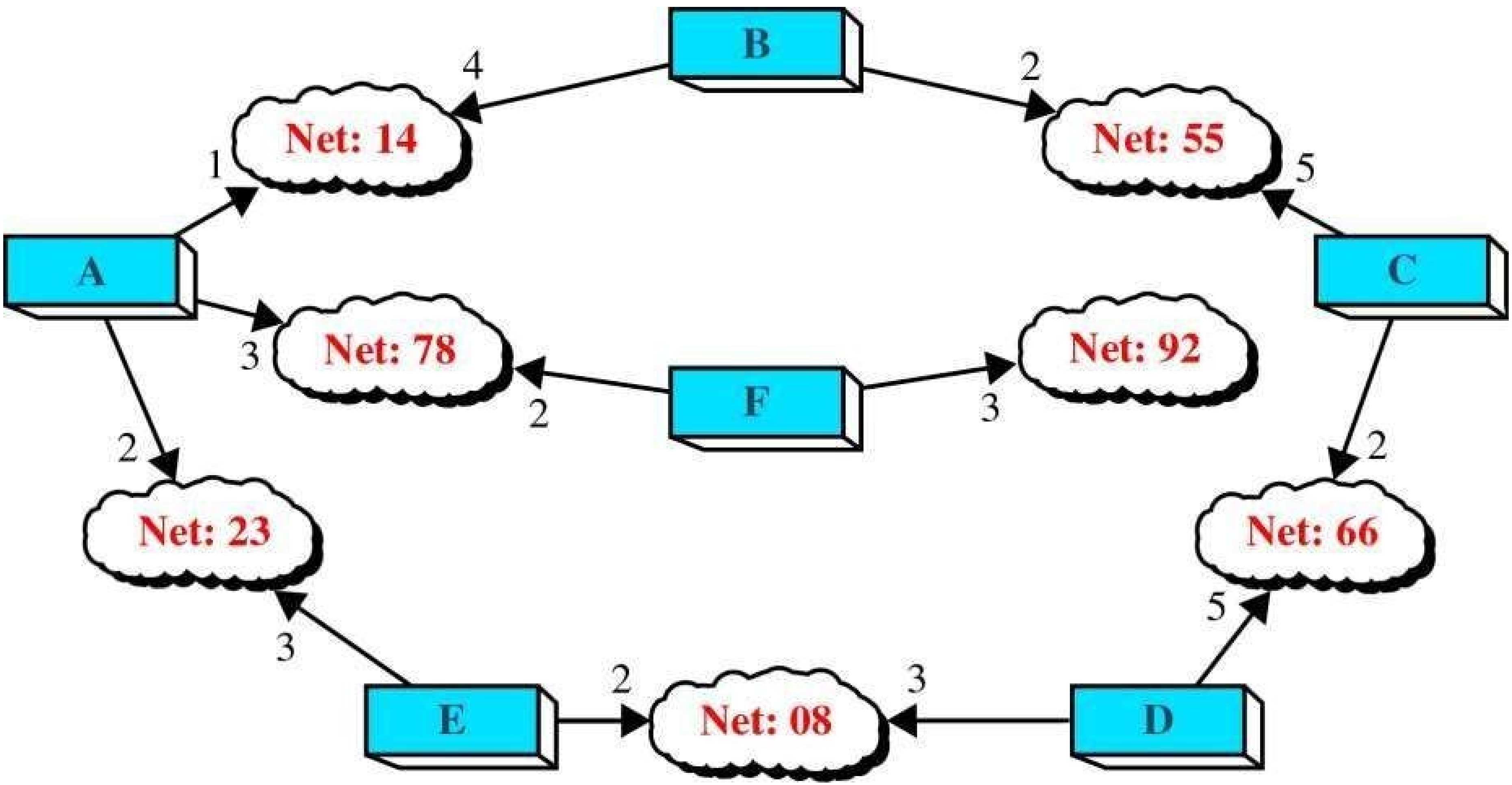
- Each router shares its knowledge about the neighborhood
- Each router sends its knowledge to all router.

  Flooding -> Each router share info to neighbor, The neighbor to its own neighbor and so on.,
- Information sharing when there is a **change.**

Concept of Link State Routing



Cost in Link State Routing



Link State Packet

Advertiser	Network	Cost	Neighbor
. . . .	. . . . . .	. . . .	. . . .
. . . .	. . . . . .	. . . .	. . . .
. . . .	. . . . . .	. . . .	. . . .

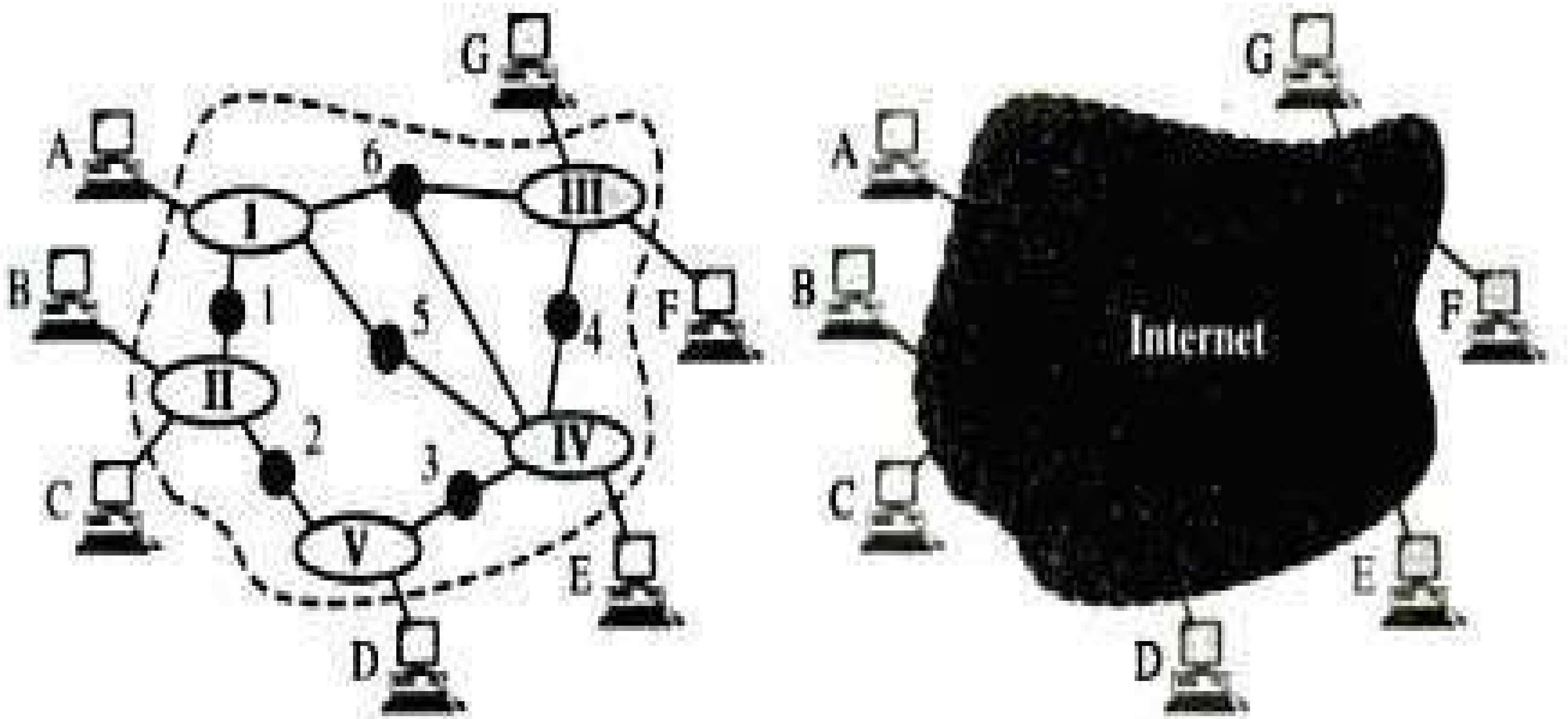
Link State Database

Advertiser	Network	Cost	Neighbor
A	14	1	B
A	78	3	F
A	23	2	E
B	14	4	A
B	55	2	C
C	55	5	B
C	66	2	D
D	66	5	C
D	08	3	E
E	23	3	A
E	08	2	D
F	78	2	A
F	92	3	—

TCP/ IP

- It was developed before OSI
- This project was funded by ARPA of U.S called ARPANET which is turned into TCP/IP
- In internet it acts like a single network connection many of any size and type.
- TCP and UDP creates a data unit called **Segment** or datagram.

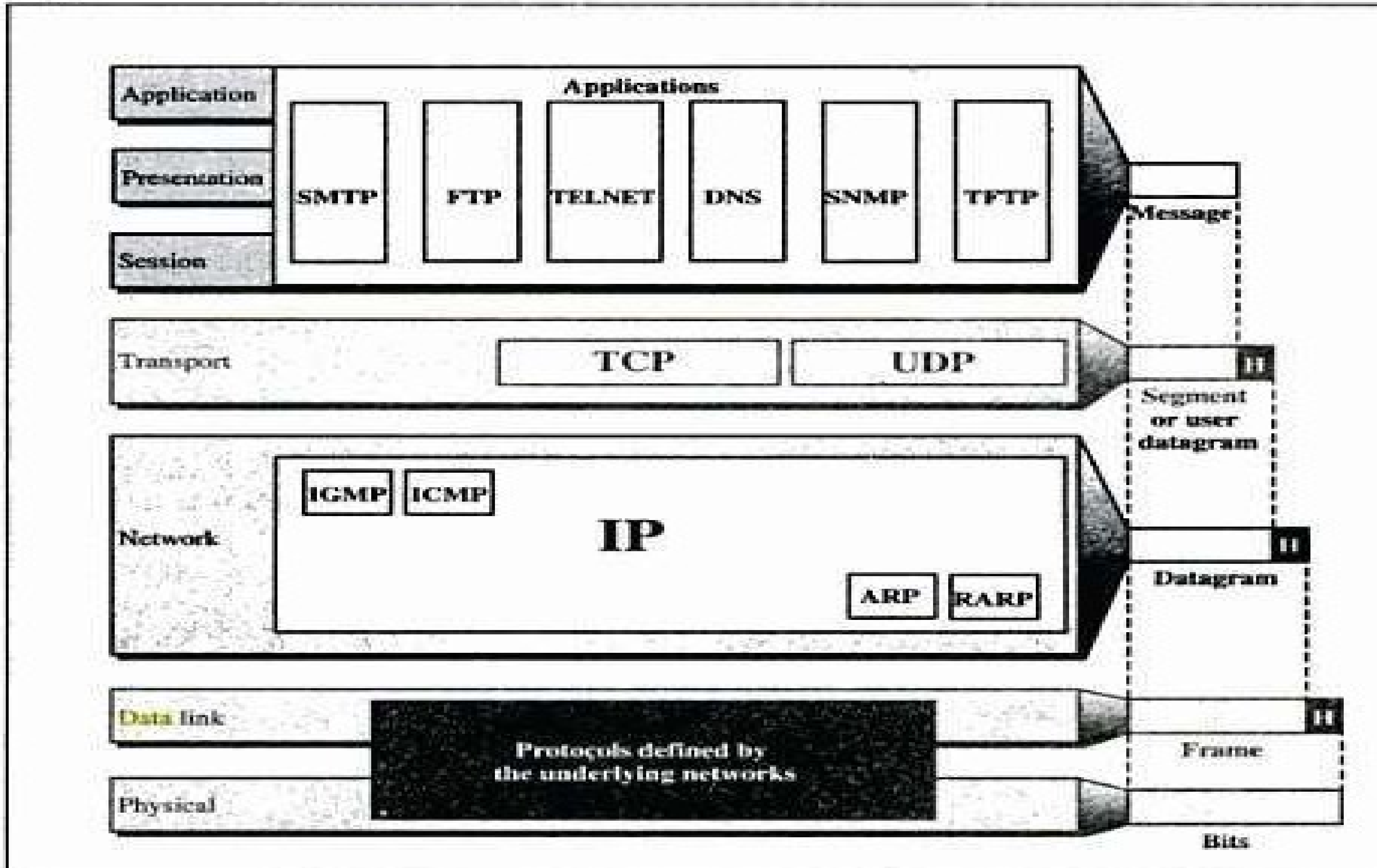
Figure 24.1 *An internet according to TCP/IP*



a. An actual internet

b. An internet seen by TCP/IP

Figure 24.2 *TCP/IP and OSI model*



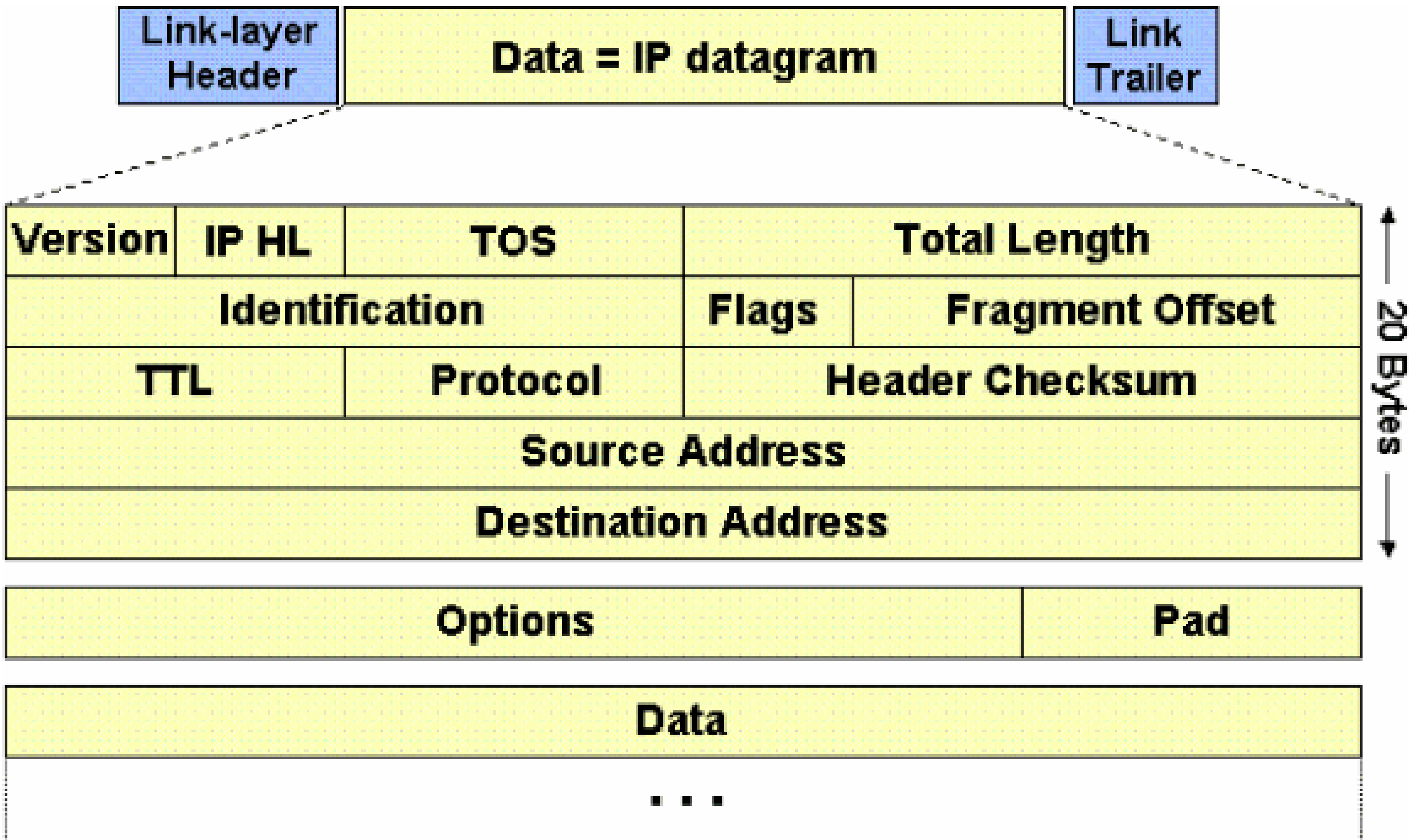
What is an IP address?

- An **Internet Protocol address** is a numerical label assigned to each device connected to a computer network that uses the Internet Protocol for communication.
- An IP address serves two principal functions: host or network interface identification and location addressing.

IP (Internet Proocol)

- Network layer of TCP/IP supports IP in turn four other supporting protocol
  - ICMP
  - IGMP
  - ARP
  - RARP
- It is a transmission mechanism used by TCP/IP protocols

IP datagram



Link-layer Header
Data = IP datagram
Link Trailer
Version
IP HL
TOS
Total Length
Identification
Flags
Fragment Offset
TTL
Protocol
Header Checksum
Source Address
Destination Address
20 Bytes
Options
Pad
Data

C ont.,

- IP Is a unreliable and connection less datagram protocol.
- No error checking or tracking.
- Data transmitted to destination but no guarantees.
- IP must be paired with TCP.

IP Addressing

- In addition to physical address (NIC), to identify each device in the network it requires
- IP address.
- Address that identify host of its network.
- ***An IP address is a 32-bit address. The IP addresses are***
- ***unique and universal.***

It Represented in a Dotted-decimal

10000000 00001011 00000011 00011111

128.11.3.31

Example 1

Change the following IP addresses from binary notation to dotted- decimal notation.

a. 10000001 00001011 00001011 11101111

b. 11111001 10011011 11111011 00001111

Solution

We replace each group of 8 bits with its equivalent decimal number and add dots for separation:

a. 129.11.11.239

b. 249.155.251.15

Example 2

Change the following IP addresses from dotted-decimal notation to binary notation.

a. 111.56.45.78

b. 75.45.34.78

Solution

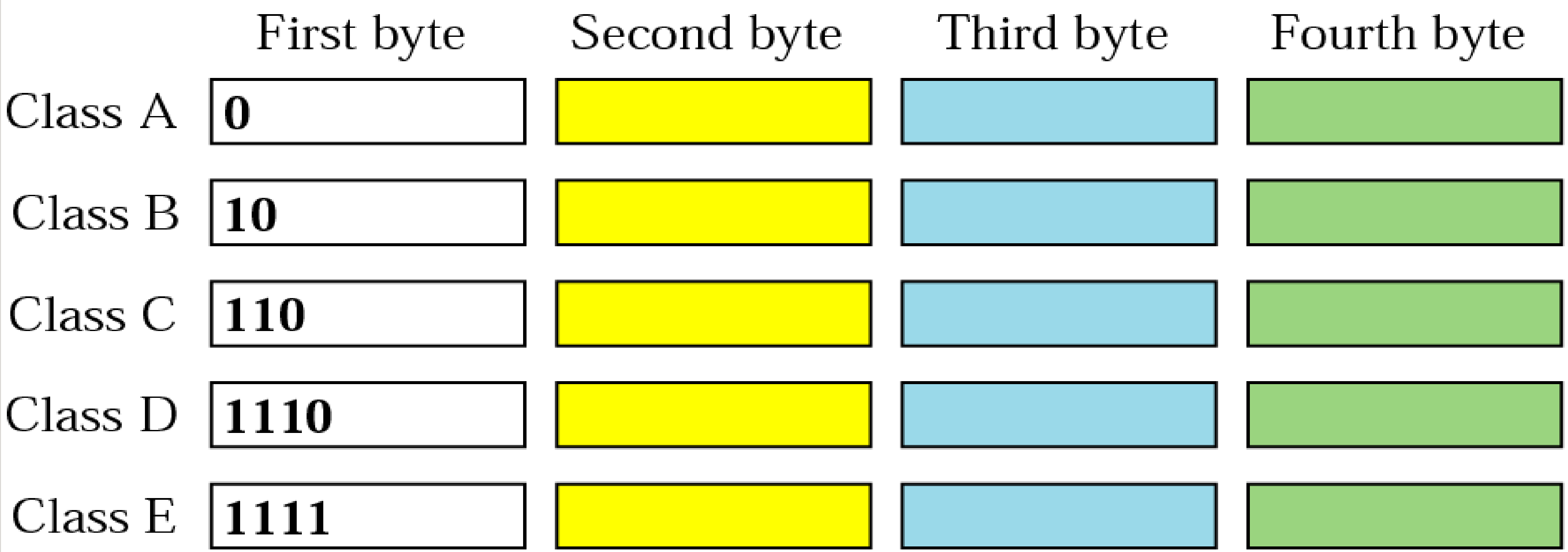
We replace each decimal number with its binary equivalent (see Appendix B):

a. 01101111 00111000 00101101 01001110
b. 01001011 00101101 00100010 01001110

Classes and Blocks

- **Class A address:** designed for large organizations with a large number of attached hosts or routers. (wasted and not used)
- **Class B address:** designed for midsize organizations with ten of thousands of attached hosts or routers(too large for many organizations)
- **Class C address:** designed for small organizations with a small number of attached hosts or routers.(too small for many organizations)
- **Class D address:** designed for multicasting. (waste of addresses)
- **Class E address:** reserved for future use (waste of addresses)

Figure 19.10 Finding the class in binary notation



Finding the class in decimal notation (changes from 0 to 255)

	First byte	Second byte	Third byte	Fourth byte
Class A	0 to 127			
Class B	128 to 191			
Class C	192 to 223			
Class D	224 to 239			
Class E	240 to 255			

Private and Public IP Address

Class	Starting IPAddress	Ending IPAddress	# of Hosts
A	10.0.0.0	10.255.255.255	16,777,216
B	172.16.0.0	172.31.255.255	1,048,576
C	192.168.0.0	192.168.255.255	65,536

Example 3

Find the class of each address:

a. **0**0000001 00001011 00001011 11101111

b. **111**10011 10011011 11111011 00001111

Solution

See the procedure in Figure 19.11.

a. **The first bit is 0; this is a class A address.**

b. **The first 4 bits are 1s; this is a class E address.**

Example 4

Find the class of each address:

a. **227**.12.14.87

b. **252**.5.15.111

c. **134**.11.78.56

Solution

a. **The first byte is 227 (between 224 and 239); the class is D.**
b. **The first byte is 252 (between 240 and 255); the class is E.**
c. **The first byte is 134 (between 128 and 191); the class is B.**

Types of IP address

- Static address
- Dynamic address
-

Static IP address

- manually input by network administrator.
- manageable for small networks

Types of IP address

- <u>Dynamic IP address</u>
- examples - BOOTP, DHCP
  - Assigned by server when host boots
  - Derived automatically from a range of addresses
  - Duration of 'lease' negotiated, then address released back to server

Subnetting

- Dividing the network into several smaller groups (subnets) with each group having its own **subnet IP address.**
- Site looks to rest of internet like **<u>single network</u>** and routers outside the organization route the packet based on the main Network address.
- Local routers route within subnetted network <u>using subnet address.</u>

Subnetting

- Host portion of address partitioned into subnet number (most significant part) and host number (least significant part)
- In this case, IP address will have **3 levels** (Main network, subnet, host)
- **Subnet mask** is a 32-bit consists of zeros and ones that indicates which bits of the IP address are subnet number and which are host number
- Subnet mask when AND ed with the IP address it gives the subnetwork address

Masking

Masking is a process that extracts the address of the physical network from an IP address.

Boundary level masking: Here the mask numbers are either 255 or 0, finding the subnetwork address is very easy.

Non-boundary level masking.

If mask numbers are not just 255 or 0, finding the subnetwork address

Supernetting:

- Supernetting combines several networks into one lager one (Because of Address reduction)

IP Network Addressing

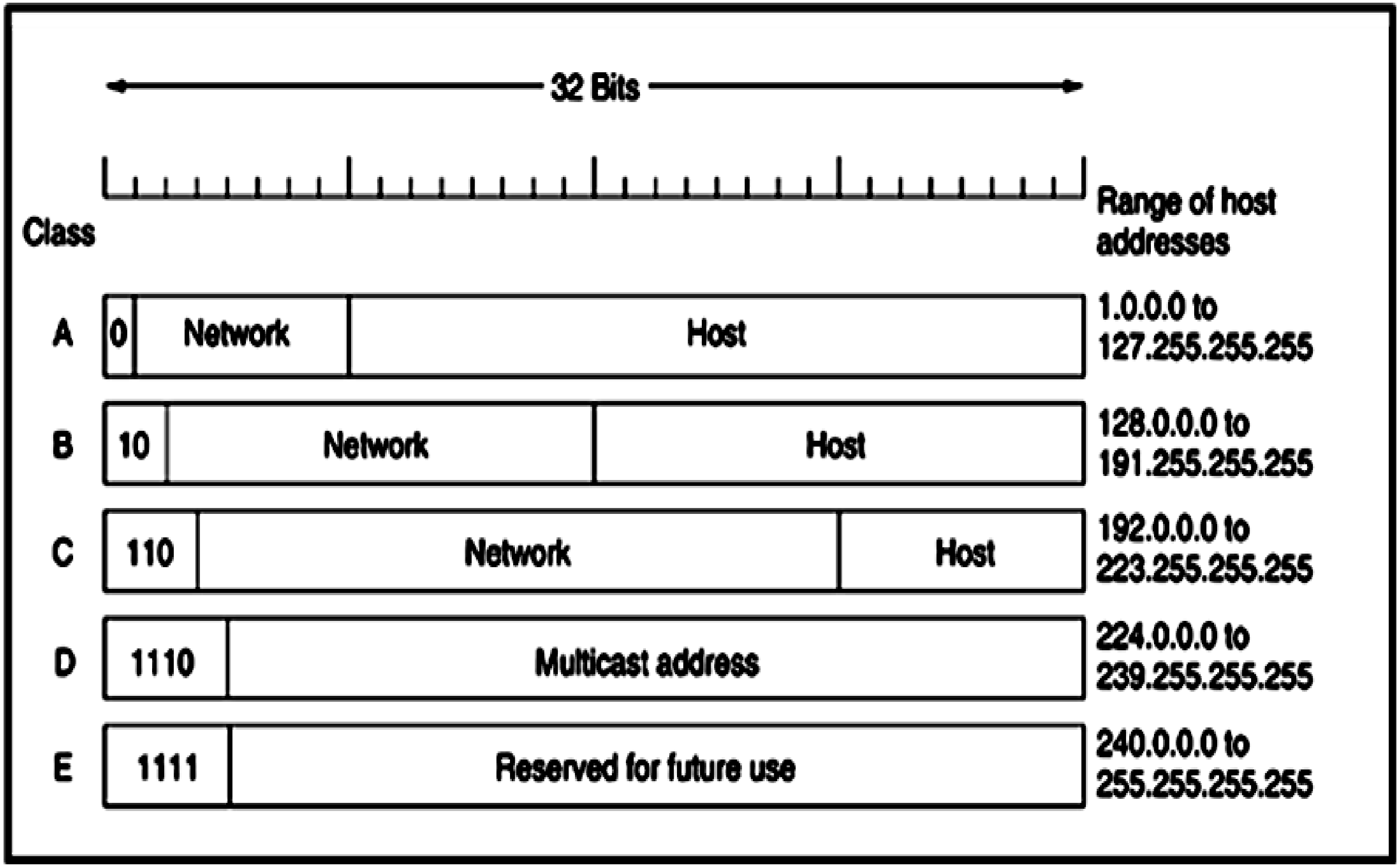
- INTERNET → world's largest public data network, doubling in size every nine months
- IPv4, defines a 32-bit address - 2^{32} (4,294,967,296) IPv4 addresses available
- The first problem is concerned with the eventual depletion of the IP address space.
- Traditional model of classful addressing does not allow the address space to be used to its maximum potential.

ClassfulAddressing

- When IP was first standardized in Sep 1981, each system attached to the IP based Internet had to be assigned a unique 32-bit address
- The 32-bit IP addressing scheme involves a two level addressing hierarchy

Network Number/Prefix	Host Number

Classful Addressing



Classless Addressing

- Classless Inter-Domain Routing (CIDR) is another name for classless addressing.
- This addressing type aids in the more efficient allocation of IP addresses.
- This technique assigns a block of IP addresses based on specified conditions when the user demands a specific amount of IP addresses.
- This block is known as a "CIDR block", and it contains the necessary number of IP addresses.

Classless Addressing

When allocating a block, classless addressing is concerned with the following three rules.

- **Rule 1** – The CIDR block's IP addresses must all be contiguous.
- **Rule 2** – The block size must be a power of two to be attractive. Furthermore, the block's size is equal to the number of IP addresses in the block.
- **Rule 3** – The block's first IP address must be divisible by the block size.

Classless Addressing

- The network component has a bit count of 27, whereas the host portion has a bit count of 5. (32-27).

- The binary representation of the address is: (00100011 . 11000000 . 10101000 . 00000001).

- (11000000.10101000.00000001.00100000) is the first IP address (assigns 0 to all host bits), that is, 192.168.1.32

- (11000000.10101000.00000001.00111111) is the most recent IP address (assigns 1 to all host bits), that is, 192.168.1.63

Difference Between Classful and Classless Addressing

- The network component has a bit count of 27, whereas the host portion has a bit count of 5. (32-27).
- The binary representation of the address is: (00100011 . 11000000 . 10101000 . 00000001).
- (11000000.10101000.00000001.00100000) is the first IP address (assigns 0 to all host bits), that is, 192.168.1.32
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Internet Protocol (IP)

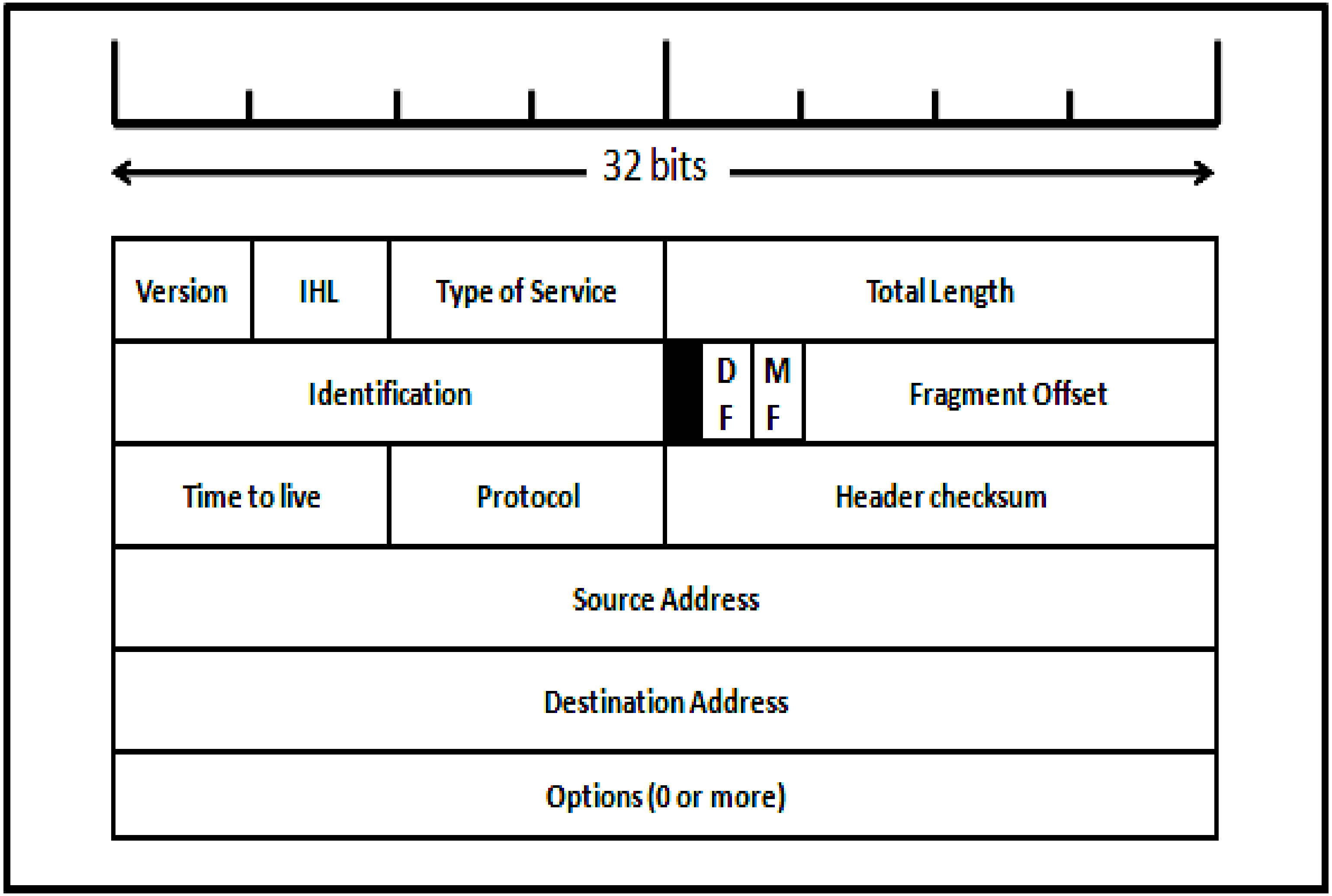
- What is Internet Protocol?
  - Internet Protocol is a set of technical rules that defines how computers communicate over a network.

  - Currently, There are two versions of IP
    - IP version 4 (IPv4)
    - IP version 6 (IPv6).

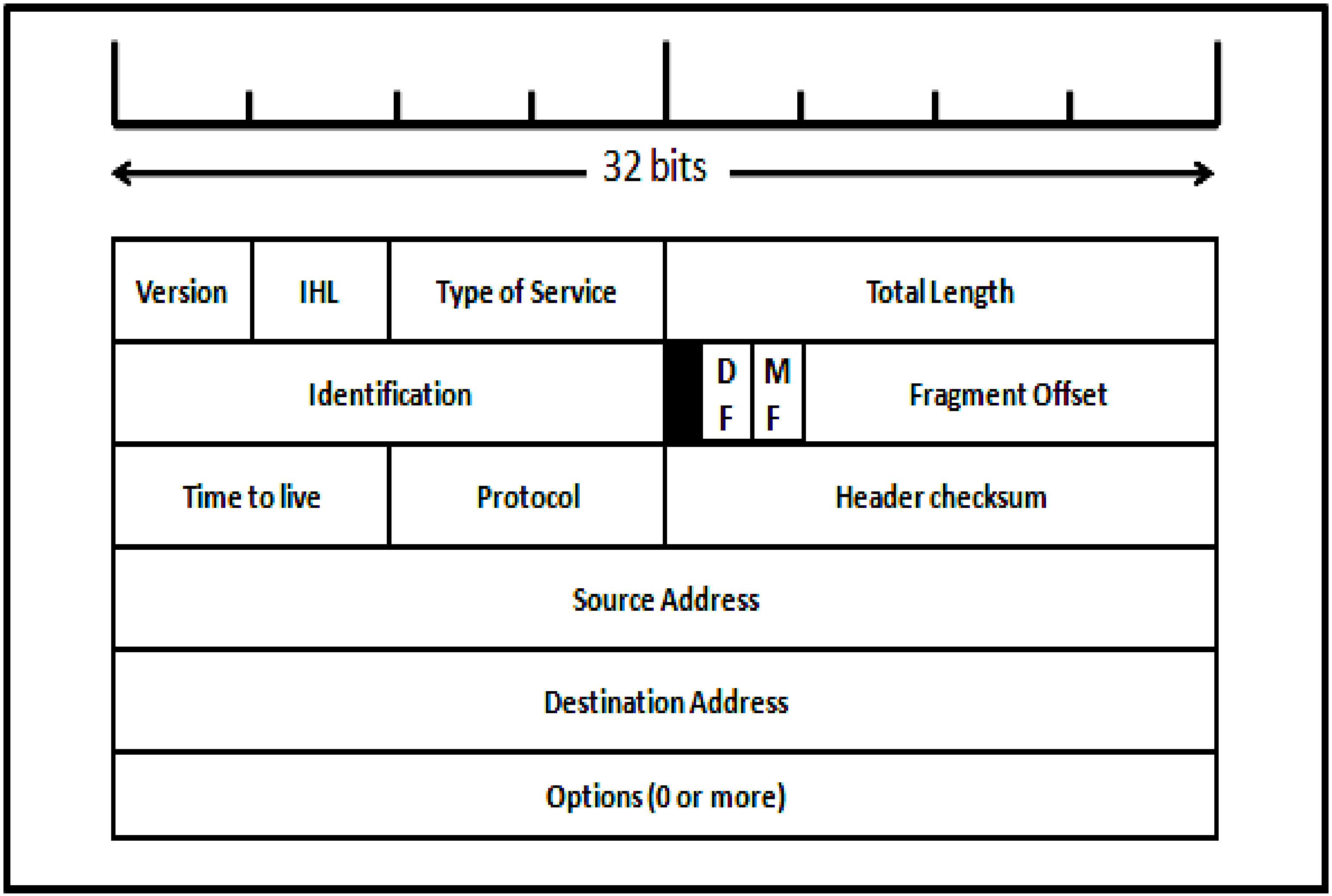
Internet Protocol (IP)

- What is IPv4?
  - IPv4 was the first version of Internet Protocol to be widely used, and accounts for most of today's Internet traffic.
  - There are just over 4 billion IPv4 addresses. While that is a lot of IP addresses, it is not enough to last forever.

Internet Protocol (IP)



Internet Protocol (IP) header



Internet Protocol (IP) header

No	Field Name	Description
1	Version	Keeps track of the version of the protocol the datagram belongs to (IPV4 or IPv6)
2	IHL	Used to indicate the length of the Header. Minimum value is 5 Maximum value 15
3	Type of service	Used to distinguish between different classes of service
4	Total length	It includes everything in the datagram–both header and data. The maximum length is 65,535 bytes
5	Identification	Used to allow the destination host to identify which datagram a newly arrived fragment belongs to. All the fragments of a datagram contain the same Identification value
6	DF	1 bit field. It stands for Don't Fragment. Signals the routers not to fragment the datagram because the destination is incapable of putting the pieces back together again
7	MF	MF stands for More Fragments. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.
8	Fragment offset	Used to determine the position of the fragment in the current datagram.

Internet Protocol (IP) header

9	Time to live	It is a counter used to limit packet lifetimes. It must be decremented on each hop. When it hits zero, the packet is discarded and a warning packet is sent back to the source host.
10	Header checksum	It verifies Header for errors.
11	Source address	IP address of the source
12	Destination address	IP address of the destination
13	Options	The options are variable length. Originally, five options were defined: 1. Security : specifies how secret the datagram is 2. Strict source routing : Gives complete path to be followed 3. Loose source routing : Gives a list of routers not to be missed 4. Record route: Makes each router append its IP address 5. Timestamp: Makes each router append its IP address and timestamp

Internet Protocol (IP)

- What is IPv6?
  - IPv6 is a newer numbering system that provides a much larger address pool than IPv4. It was deployed in 1999 and should meet the world’s IP addressing needs well into the future.

Internet Protocol (IP)

- What is the major difference?
  - The major difference between IPv4 and IPv6 is the number of IP addresses.

  - There are 4,294,967,296 IPv4 addresses.
  - while, there are 340,282,366,920,938,463,463,374,607,431,768,211,456 IPv6 addresses.

128-bit IPv6 Address

128-bit IPv6 Address

3FFE:085B:1F1F:0000:0000:0000:00A9:1234

8 groups of 16-bit hexadecimal numbers separated by “:”

Leading zeros can be removed

3FFE:85B:1F1F::A9:1234

:: = all zeros in one or more group of 16-bit hexadecimal numbers

IPv4 vs. IPv6

	IPv6
IPv4 addresses are 32 bit length.	IPv6 addresses are 128 bit length.
IPv4 addresses are binary numbers represented in decimals.	IPv6 addresses are binary numbers represented in hexadecimals.
IPSec support is only optional.	Inbuilt IPSec support.
Fragmentation is done by Sender and forwarding routers.	Fragmentation is done only by sender.

No packet flow identification.	Packet flow identification is available within the IPv6 header using the Flow Label field.
Checksum field is available in IPv4 header	No checksum field in IPv6 header.
Options fields are available in IPv4 header.	No option fields, but IPv6 Extension headers are available.
Address Resolution Protocol (ARP) is available to map IPv4 addresses to MAC addresses.	Address Resolution Protocol (ARP) is replaced with a function of Neighbor Discovery Protocol (NDP).

Internet Group Management Protocol (IGMP) is used to manage multicast group membership.	IGMP is replaced with Multicast Listener Discovery (MLD) messages.
Broadcast messages are available.	Broadcast messages are not available. Instead a link- local scope "All nodes" multicast IPv6 address(FF02::1) is used for broadcast similar functionality.
Manual configuration (Static) of IPv4 addresses or DHCP (Dynamic Host configuration Protocol)	Auto-configuration of addresses is available.

Why IPv6?

- Deficiency of IPv4
- Address space
- exhaustion

New types of service → Integration

- Multicast
- Quality of Service
- Security
- Mobility (MIPv6)

Header and format

Advantages of IPv6 over IPv4

- Larger address
- space Better header
- format New options
- Allowance for extension
- Support for resource
- allocation Support for more
- security Support for mobility

THANK YOU